Autograder Results

Results | Code

0.0) test_submitted_files (test_file_exists.TestFiles) (10/10)

```
Test: Checking Exists: proj.py
Success!
```

1) test_1 (test_generated_cases.simplebtree) (10/10)

```
Test Set 1: Performing 100 tests with size = 10 and type = insert then
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
initialize,8
insert,23
insert,58
insert,21
insert,17
insert,70
insert,51
insert,87
insert,84
insert,57
insert,3
dump
---> Your Solution:
- keys = [51]
  children =
  - keys = [3, 17, 21, 23]
    parent keys = [51]
    children = None
  - keys = [57, 58, 70, 84, 87]
    parent keys = [51]
    children = None

---> Correct Solution:
- keys = [51]
  children =
  - keys = [3, 17, 21, 23]
    parent keys = [51]
    children = None
  - keys = [57, 58, 70, 84, 87]
    parent keys = [51]
    children = None

Sub-Tests #2-#100: Passed (Detailed Omitted)
Test Set 1: Passed Completely!
```

2) test_2 (test_generated_cases.simplebtree) (10/10)

```
Test Set 2: Performing 100 tests with size = 10 and type = insert and d
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
initialize,4
insert,6
insert,97
insert,93
insert,23
insert,63
insert,5
insert,32
insert,28
insert,80
delete,63
delete,23
delete,28
delete,32
dump
---> Your Solution:
- keys = [6, 93]
  children =
  - keys = [5]
    parent keys = [6, 93]
    children = None
  - keys = [80]
    parent keys = [6, 93]
    children = None
  - keys = [97]
    parent keys = [6, 93]
    children = None

---> Correct Solution:
- keys = [6, 93]
  children =
  - keys = [5]
    parent keys = [6, 93]
    children = None
  - keys = [80]
    parent keys = [6, 93]
    children = None
  - keys = [97]
    parent keys = [6, 93]
    children = None

Sub-Tests #2-#100: Passed (Detailed Omitted)
Test Set 2: Passed Completely!
```

3) test_3 (test_generated_cases.simplebtree) (10/10)

```
Test Set 3: Performing 100 tests with size = 10 and type = insert and d
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
initialize,9
insert,22
insert,43
insert,81
insert,61
insert,33
insert,29
delete,81
delete,61
delete,43
insert,18
insert,88
insert,32
search,22
---> Your Solution:
[]

---> Correct Solution:
[]

Sub-Tests #2-#100: Passed (Detailed Omitted)
Test Set 3: Passed Completely!
```

4) test_4 (test_generated_cases.simplebtree) (10/10)

```
Test Set 4: Performing 10 tests with size = 100 and type = insert then
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
initialize,6
insert,219
insert,661
insert,153
insert,95
insert,215
insert,375
insert,623
insert,446
insert,270
insert,755
insert,466
insert,311
insert,162
insert,131
insert,88
insert,508
insert,350
insert,26
insert,943
insert,328
insert,992
insert,875
insert,130
insert,898
insert,608
insert,171
insert,264
insert,871
insert,988
insert,798
insert,640
insert,456
insert,424
insert,861
insert,993
insert,474
insert,81
insert,289
insert,485
insert,45
insert,9
insert,137
insert,272
insert,413
```

```
insert,261
insert,711
insert,590
insert,563
insert,809
insert,945
insert,92
insert,266
insert,748
insert,973
insert,25
insert,102
insert,997
insert,242
insert,186
insert,8
insert,979
insert,230
insert,94
insert,15
insert,813
insert,155
insert,354
insert,638
insert,842
insert,487
insert,150
insert,514
insert,500
insert,251
insert,574
insert,598
insert,238
insert,420
insert,278
insert,178
insert,435
insert,542
insert,228
insert,583
insert,263
insert,60
insert,641
insert,65
insert,198
insert,999
insert,545
insert,151
insert,212
```

```
insert,239
insert,933
insert,569
insert,318
insert,759
insert,20
insert,554
dump
---> Your Solution:
- keys = [137, 272, 598]
  children =
  - keys = [15, 45, 94]
    parent keys = [137, 272, 598]
    children =
    - keys = [8, 9]
      parent keys = [15, 45, 94]
      children = None
    - keys = [20, 25, 26]
      parent keys = [15, 45, 94]
      children = None
    - keys = [60, 65, 81, 88, 92]
      parent keys = [15, 45, 94]
      children = None
    - keys = [95, 102, 130, 131]
      parent keys = [15, 45, 94]
      children = None
  - keys = [171, 219, 251]
    parent keys = [137, 272, 598]
    children =
    - keys = [150, 151, 153, 155, 162]
      parent keys = [171, 219, 251]
      children = None
    - keys = [178, 186, 198, 212, 215]
      parent keys = [171, 219, 251]
      children = None
    - keys = [228, 230, 238, 239, 242]
      parent keys = [171, 219, 251]
      children = None
    - keys = [261, 263, 264, 266, 270]
      parent keys = [171, 219, 251]
      children = None
  - keys = [311, 375, 456, 508, 563]
    parent keys = [137, 272, 598]
    children =
    - keys = [278, 289]
      parent keys = [311, 375, 456, 508, 563]
      children = None
    - keys = [318, 328, 350, 354]
      parent keys = [311, 375, 456, 508, 563]
```

```
      children = None
    - keys = [413, 420, 424, 435, 446]
      parent keys = [311, 375, 456, 508, 563]
      children = None
    - keys = [466, 474, 485, 487, 500]
      parent keys = [311, 375, 456, 508, 563]
      children = None
    - keys = [514, 542, 545, 554]
      parent keys = [311, 375, 456, 508, 563]
      children = None
    - keys = [569, 574, 583, 590]
      parent keys = [311, 375, 456, 508, 563]
      children = None
  - keys = [661, 798, 861, 945, 992]
    parent keys = [137, 272, 598]
    children =
    - keys = [608, 623, 638, 640, 641]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [711, 748, 755, 759]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [809, 813, 842]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [871, 875, 898, 933, 943]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [973, 979, 988]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [993, 997, 999]
      parent keys = [661, 798, 861, 945, 992]
      children = None

---> Correct Solution:
- keys = [137, 272, 598]
  children =
  - keys = [15, 45, 94]
    parent keys = [137, 272, 598]
    children =
    - keys = [8, 9]
      parent keys = [15, 45, 94]
      children = None
    - keys = [20, 25, 26]
      parent keys = [15, 45, 94]
      children = None
    - keys = [60, 65, 81, 88, 92]
      parent keys = [15, 45, 94]
```

```
    children = None
  - keys = [95, 102, 130, 131]
    parent keys = [15, 45, 94]
    children = None
- keys = [171, 219, 251]
  parent keys = [137, 272, 598]
  children =
  - keys = [150, 151, 153, 155, 162]
    parent keys = [171, 219, 251]
    children = None
  - keys = [178, 186, 198, 212, 215]
    parent keys = [171, 219, 251]
    children = None
  - keys = [228, 230, 238, 239, 242]
    parent keys = [171, 219, 251]
    children = None
  - keys = [261, 263, 264, 266, 270]
    parent keys = [171, 219, 251]
    children = None
- keys = [311, 375, 456, 508, 563]
  parent keys = [137, 272, 598]
  children =
  - keys = [278, 289]
    parent keys = [311, 375, 456, 508, 563]
    children = None
  - keys = [318, 328, 350, 354]
    parent keys = [311, 375, 456, 508, 563]
    children = None
  - keys = [413, 420, 424, 435, 446]
    parent keys = [311, 375, 456, 508, 563]
    children = None
  - keys = [466, 474, 485, 487, 500]
    parent keys = [311, 375, 456, 508, 563]
    children = None
  - keys = [514, 542, 545, 554]
    parent keys = [311, 375, 456, 508, 563]
    children = None
  - keys = [569, 574, 583, 590]
    parent keys = [311, 375, 456, 508, 563]
    children = None
- keys = [661, 798, 861, 945, 992]
  parent keys = [137, 272, 598]
  children =
  - keys = [608, 623, 638, 640, 641]
    parent keys = [661, 798, 861, 945, 992]
    children = None
  - keys = [711, 748, 755, 759]
    parent keys = [661, 798, 861, 945, 992]
    children = None
```

```
    - keys = [809, 813, 842]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [871, 875, 898, 933, 943]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [973, 979, 988]
      parent keys = [661, 798, 861, 945, 992]
      children = None
    - keys = [993, 997, 999]
      parent keys = [661, 798, 861, 945, 992]
      children = None

Sub-Tests #2-#10: Passed (Detailed Omitted)
```

5) test_5 (test_generated_cases.simplebtree) (0/10)

```
Test Set 5: Performing 10 tests with size = 100 and type = insert and d
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,9
insert,638
insert,644
.
.
.
insert,501
insert,792
dump
---> Your Solution:
- keys = [56, 338, 507, 690, 856]
  children =
  - keys = [25, 31, 39, 49]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [120, 171, 185, 234, 250]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [400, 414, 433, 464, 466, 475, 493, 501]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [516, 532, 586, 591, 611, 633, 652]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [706, 721, 727, 731, 792, 814, 822, 850]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [862, 867, 887, 893, 932, 937, 947, 987]
    parent keys = [56, 338, 507, 690, 856]
    children = None

---> Correct Solution:
- keys = [56, 338, 507, 690, 856]
  children =
  - keys = [25, 31, 39, 49]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [120, 171, 185, 234, 250]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [400, 414, 433, 464, 466, 475, 493, 501]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [516, 532, 586, 591, 611, 633, 652]
```

```
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [706, 721, 727, 731, 792, 814, 822, 850]
    parent keys = [56, 338, 507, 690, 856]
    children = None
  - keys = [862, 867, 887, 893, 932, 937, 947, 987]
    parent keys = [56, 338, 507, 690, 856]
    children = None

Sub-Test #2-#3: Passed (Details Omitted)
Sub-Test #4: Failed (Details follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,9
insert,463
insert,465
.
.
.
insert,533
insert,375
dump
---> Your Solution:
- keys = [106, 151, 256, 418, 812]
  children =
  - keys = [14, 17, 23, 28, 31, 79, 93, 96]
    parent keys = [106, 151, 256, 418, 812]
    children = None
  - keys = [117, 128, 133, 134, 144]
    parent keys = [106, 151, 256, 418, 812]
    children = None
  - keys = [152, 160, 175, 180, 206, 227]
    parent keys = [106, 151, 256, 418, 812]
    children = None
  - keys = [296, 316, 319, 325, 331, 353, 375, 385]
    parent keys = [106, 151, 256, 418, 812]
    children = None
  - keys = [452, 533, 540, 563, 650, 653, 724, 779]
    parent keys = [106, 151, 256, 418, 812]
    children = None
  - keys = [939, 949, 964, 967, 969, 977, 983, 992]
    parent keys = [106, 151, 256, 418, 812]
    children = None

---> Correct Solution:
- keys = [79, 133, 227, 353, 653, 967]
  children =
  - keys = [14, 17, 23, 28, 31]
    parent keys = [79, 133, 227, 353, 653, 967]
```

```
    children = None
  - keys = [93, 96, 106, 117, 128]
    parent keys = [79, 133, 227, 353, 653, 967]
    children = None
  - keys = [134, 144, 151, 152, 160, 175, 180, 206]
    parent keys = [79, 133, 227, 353, 653, 967]
    children = None
  - keys = [256, 296, 316, 319, 325, 331]
    parent keys = [79, 133, 227, 353, 653, 967]
    children = None
  - keys = [375, 385, 418, 452, 533, 540, 563, 650]
    parent keys = [79, 133, 227, 353, 653, 967]
    children = None
  - keys = [724, 779, 812, 939, 949, 964]
    parent keys = [79, 133, 227, 353, 653, 967]
    children = None
  - keys = [969, 977, 983, 992]
    parent keys = [79, 133, 227, 353, 653, 967]
    children = None

Note: Once a Sub-Test fails, no more Sub-Tests in that Test Set are run
Test Set 5: Failed!
```

6) test_6 (test_generated_cases.simplebtree) (10/10)

```
Test Set 6: Performing 10 tests with size = 100 and type = insert and d
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,9
insert,680
insert,948
.
.
.
insert,721
insert,80
search,18
---> Your Solution:
[0]

---> Correct Solution:
[0]

Sub-Tests #2-#10: Passed (Detailed Omitted)
Test Set 6: Passed Completely!
```

7) test_7 (test_generated_cases.simplebtree) (10/10)

```
Test Set 7: Performing 5 tests with size = 1000 and type = insert then
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,3
insert,4772
insert,88
.
.
.
insert,2981
insert,3452
dump
---> Your Solution:
- keys = [2542, 6923]
  children =
  - keys = [876]
    parent keys = [2542, 6923]
    children =
    - keys = [482]
      parent keys = [876]
      children =
      - keys = [154, 296]
        parent keys = [482]
        children =
        - keys = [31, 88]
          parent keys = [154, 296]
          children =
          - keys = [12]
            parent keys = [31, 88]
            children =
            - keys = [1, 11]
              parent keys = [12]
              children = None
            - keys = [19, 25]
              parent keys = [12]
              children = None
          - keys = [61]
            parent keys = [31, 88]
            children =
            - keys = [47]
              parent keys = [61]
              children = None
            - keys = [82]
              parent keys = [61]
              children = None
          - keys = [116, 128]
            parent keys = [31, 88]
```

```
      children =
      - keys = [89, 106]
        parent keys = [116, 128]
        children = None
      - keys = [124]
        parent keys = [116, 128]
        children = None
      - keys = [133, 145]
        parent keys = [116, 128]
        children = None
  - keys = [180, 258]
    parent keys = [154, 296]
    children =
    - keys = [172, 177]
      parent keys = [180, 258]
      children =
      - keys = [156, 161]
        parent keys = [172, 177]
        children = None
      - keys = [175]
        parent keys = [172, 177]
        children = None
      - keys = [178]
        parent keys = [172, 177]
        children = None
    - keys = [210]
      parent keys = [180, 258]
      children =
      - keys = [185, 204]
        parent keys = [210]
        children = None
      - keys = [228, 256]
        parent keys = [210]
        children = None
    - keys = [270, 284]
      parent keys = [180, 258]
      children =
      - keys = [267, 268]
        parent keys = [270, 284]
        children = None
      - keys = [272, 283]
        parent keys = [270, 284]
        children = None
      - keys = [292]
        parent keys = [270, 284]
        children = None
  - keys = [351, 380]
    parent keys = [154, 296]
    children =
```

```
        - keys = [329]
          parent keys = [351, 380]
          children =
          - keys = [301, 307]
            parent keys = [329]
            children = None
          - keys = [339]
            parent keys = [329]
            children = None
        - keys = [373]
          parent keys = [351, 380]
          children =
          - keys = [364, 369]
            parent keys = [373]
            children = None
          - keys = [374, 377]
            parent keys = [373]
            children = None
        - keys = [425, 450]
          parent keys = [351, 380]
          children =
          - keys = [408]
            parent keys = [425, 450]
            children = None
          - keys = [443, 446]
            parent keys = [425, 450]
            children = None
          - keys = [462, 481]
            parent keys = [425, 450]
            children = None
    - keys = [583, 712]
      parent keys = [482]
      children =
      - keys = [543]
        parent keys = [583, 712]
        children =
        - keys = [500, 538]
          parent keys = [543]
          children =
          - keys = [486, 493]
            parent keys = [500, 538]
            children = None
          - keys = [517]
            parent keys = [500, 538]
            children = None
          - keys = [541]
            parent keys = [500, 538]
            children = None
        - keys = [568, 580]
```

```
      parent keys = [543]
      children =
      - keys = [567]
        parent keys = [568, 580]
        children = None
      - keys = [570, 577]
        parent keys = [568, 580]
        children = None
      - keys = [581, 582]
        parent keys = [568, 580]
        children = None
  - keys = [623, 652]
    parent keys = [583, 712]
    children =
    - keys = [601, 619]
      parent keys = [623, 652]
      children =
      - keys = [590, 599]
        parent keys = [601, 619]
        children = None
      - keys = [609]
        parent keys = [601, 619]
        children = None
      - keys = [622]
        parent keys = [601, 619]
        children = None
    - keys = [631]
      parent keys = [623, 652]
      children =
      - keys = [628]
        parent keys = [631]
        children = None
      - keys = [646]
        parent keys = [631]
        children = None
    - keys = [674, 702]
      parent keys = [623, 652]
      children =
      - keys = [672]
        parent keys = [674, 702]
        children = None
      - keys = [698]
        parent keys = [674, 702]
        children = None
      - keys = [705, 706]
        parent keys = [674, 702]
        children = None
  - keys = [754, 798]
    parent keys = [583, 712]
```

```
        children =
        - keys = [746]
          parent keys = [754, 798]
          children =
          - keys = [721, 729]
            parent keys = [746]
            children = None
          - keys = [747]
            parent keys = [746]
            children = None
        - keys = [759]
          parent keys = [754, 798]
          children =
          - keys = [756]
            parent keys = [759]
            children = None
          - keys = [762, 786]
            parent keys = [759]
            children = None
        - keys = [855, 867]
          parent keys = [754, 798]
          children =
          - keys = [818, 853]
            parent keys = [855, 867]
            children = None
          - keys = [857]
            parent keys = [855, 867]
            children = None
          - keys = [871, 874]
            parent keys = [855, 867]
            children = None
- keys = [1500, 1959]
  parent keys = [876]
  children =
  - keys = [1124, 1385]
    parent keys = [1500, 1959]
    children =
    - keys = [917, 1045]
      parent keys = [1124, 1385]
      children =
      - keys = [905]
        parent keys = [917, 1045]
        children =
        - keys = [895, 899]
          parent keys = [905]
          children = None
        - keys = [913, 915]
          parent keys = [905]
          children = None
```

```
      - keys = [964, 1023]
        parent keys = [917, 1045]
        children =
        - keys = [934]
          parent keys = [964, 1023]
          children = None
        - keys = [968, 1008]
          parent keys = [964, 1023]
          children = None
        - keys = [1029, 1044]
          parent keys = [964, 1023]
          children = None
      - keys = [1074, 1092]
        parent keys = [917, 1045]
        children =
        - keys = [1046, 1054]
          parent keys = [1074, 1092]
          children = None
        - keys = [1076, 1077]
          parent keys = [1074, 1092]
          children = None
        - keys = [1102, 1117]
          parent keys = [1074, 1092]
          children = None
    - keys = [1236, 1287]
      parent keys = [1124, 1385]
      children =
      - keys = [1171, 1220]
        parent keys = [1236, 1287]
        children =
        - keys = [1139, 1148]
          parent keys = [1171, 1220]
          children = None
        - keys = [1174, 1197]
          parent keys = [1171, 1220]
          children = None
        - keys = [1224, 1230]
          parent keys = [1171, 1220]
          children = None
      - keys = [1257]
        parent keys = [1236, 1287]
        children =
        - keys = [1239, 1242]
          parent keys = [1257]
          children = None
        - keys = [1260]
          parent keys = [1257]
          children = None
      - keys = [1319, 1358]
```

```
          parent keys = [1236, 1287]
          children =
          - keys = [1288, 1292]
            parent keys = [1319, 1358]
            children = None
          - keys = [1339, 1347]
            parent keys = [1319, 1358]
            children = None
          - keys = [1361, 1376]
            parent keys = [1319, 1358]
            children = None
      - keys = [1426]
        parent keys = [1124, 1385]
        children =
        - keys = [1393]
          parent keys = [1426]
          children =
          - keys = [1386, 1391]
            parent keys = [1393]
            children = None
          - keys = [1416, 1422]
            parent keys = [1393]
            children = None
        - keys = [1446, 1458]
          parent keys = [1426]
          children =
          - keys = [1432, 1434]
            parent keys = [1446, 1458]
            children = None
          - keys = [1451]
            parent keys = [1446, 1458]
            children = None
          - keys = [1478]
            parent keys = [1446, 1458]
            children = None
    - keys = [1638, 1754]
      parent keys = [1500, 1959]
      children =
      - keys = [1570]
        parent keys = [1638, 1754]
        children =
        - keys = [1538]
          parent keys = [1570]
          children =
          - keys = [1503, 1532]
            parent keys = [1538]
            children = None
          - keys = [1550, 1561]
            parent keys = [1538]
```

```
      children = None
  - keys = [1592, 1606]
    parent keys = [1570]
    children =
    - keys = [1575, 1581]
      parent keys = [1592, 1606]
      children = None
    - keys = [1598]
      parent keys = [1592, 1606]
      children = None
    - keys = [1614, 1625]
      parent keys = [1592, 1606]
      children = None
- keys = [1712]
  parent keys = [1638, 1754]
  children =
  - keys = [1685]
    parent keys = [1712]
    children =
    - keys = [1659, 1681]
      parent keys = [1685]
      children = None
    - keys = [1692, 1694]
      parent keys = [1685]
      children = None
  - keys = [1730]
    parent keys = [1712]
    children =
    - keys = [1718, 1727]
      parent keys = [1730]
      children = None
    - keys = [1752]
      parent keys = [1730]
      children = None
- keys = [1806, 1862]
  parent keys = [1638, 1754]
  children =
  - keys = [1780]
    parent keys = [1806, 1862]
    children =
    - keys = [1764, 1771]
      parent keys = [1780]
      children = None
    - keys = [1800]
      parent keys = [1780]
      children = None
  - keys = [1820, 1836]
    parent keys = [1806, 1862]
    children =
```

```
        - keys = [1810]
          parent keys = [1820, 1836]
          children = None
        - keys = [1825, 1830]
          parent keys = [1820, 1836]
          children = None
        - keys = [1849, 1861]
          parent keys = [1820, 1836]
          children = None
      - keys = [1867, 1915]
        parent keys = [1806, 1862]
        children =
        - keys = [1866]
          parent keys = [1867, 1915]
          children = None
        - keys = [1892, 1912]
          parent keys = [1867, 1915]
          children = None
        - keys = [1946, 1955]
          parent keys = [1867, 1915]
          children = None
- keys = [2096, 2304]
  parent keys = [1500, 1959]
  children =
  - keys = [2042]
    parent keys = [2096, 2304]
    children =
    - keys = [2011]
      parent keys = [2042]
      children =
      - keys = [1963, 1990]
        parent keys = [2011]
        children = None
      - keys = [2012, 2028]
        parent keys = [2011]
        children = None
    - keys = [2060]
      parent keys = [2042]
      children =
      - keys = [2056]
        parent keys = [2060]
        children = None
      - keys = [2071]
        parent keys = [2060]
        children = None
  - keys = [2160, 2253]
    parent keys = [2096, 2304]
    children =
    - keys = [2118, 2144]
```

```
      parent keys = [2160, 2253]
      children =
      - keys = [2105]
        parent keys = [2118, 2144]
        children = None
      - keys = [2128, 2134]
        parent keys = [2118, 2144]
        children = None
      - keys = [2151, 2157]
        parent keys = [2118, 2144]
        children = None
    - keys = [2202, 2212]
      parent keys = [2160, 2253]
      children =
      - keys = [2188, 2195]
        parent keys = [2202, 2212]
        children = None
      - keys = [2211]
        parent keys = [2202, 2212]
        children = None
      - keys = [2216, 2232]
        parent keys = [2202, 2212]
        children = None
    - keys = [2273, 2284]
      parent keys = [2160, 2253]
      children =
      - keys = [2259, 2271]
        parent keys = [2273, 2284]
        children = None
      - keys = [2283]
        parent keys = [2273, 2284]
        children = None
      - keys = [2296, 2299]
        parent keys = [2273, 2284]
        children = None
  - keys = [2355, 2475]
    parent keys = [2096, 2304]
    children =
    - keys = [2315]
      parent keys = [2355, 2475]
      children =
      - keys = [2309]
        parent keys = [2315]
        children = None
      - keys = [2332, 2334]
        parent keys = [2315]
        children = None
    - keys = [2408]
      parent keys = [2355, 2475]
```

```
            children =
            - keys = [2384]
              parent keys = [2408]
              children = None
            - keys = [2420]
              parent keys = [2408]
              children = None
          - keys = [2493, 2523]
            parent keys = [2355, 2475]
            children =
            - keys = [2483, 2487]
              parent keys = [2493, 2523]
              children = None
            - keys = [2494]
              parent keys = [2493, 2523]
              children = None
            - keys = [2540]
              parent keys = [2493, 2523]
              children = None
- keys = [4269, 5687]
  parent keys = [2542, 6923]
  children =
  - keys = [3205, 3725]
    parent keys = [4269, 5687]
    children =
    - keys = [2748, 2937]
      parent keys = [3205, 3725]
      children =
      - keys = [2606, 2654]
        parent keys = [2748, 2937]
        children =
        - keys = [2583]
          parent keys = [2606, 2654]
          children =
          - keys = [2576]
            parent keys = [2583]
            children = None
          - keys = [2600]
            parent keys = [2583]
            children = None
        - keys = [2636]
          parent keys = [2606, 2654]
          children =
          - keys = [2611, 2618]
            parent keys = [2636]
            children = None
          - keys = [2638, 2645]
            parent keys = [2636]
            children = None
```

```
    - keys = [2697, 2718]
      parent keys = [2606, 2654]
      children =
      - keys = [2660, 2696]
        parent keys = [2697, 2718]
        children = None
      - keys = [2698, 2714]
        parent keys = [2697, 2718]
        children = None
      - keys = [2734]
        parent keys = [2697, 2718]
        children = None
  - keys = [2833, 2880]
    parent keys = [2748, 2937]
    children =
    - keys = [2769]
      parent keys = [2833, 2880]
      children =
      - keys = [2758, 2764]
        parent keys = [2769]
        children = None
      - keys = [2773, 2781]
        parent keys = [2769]
        children = None
    - keys = [2856]
      parent keys = [2833, 2880]
      children =
      - keys = [2846]
        parent keys = [2856]
        children = None
      - keys = [2857]
        parent keys = [2856]
        children = None
    - keys = [2916, 2934]
      parent keys = [2833, 2880]
      children =
      - keys = [2896, 2897]
        parent keys = [2916, 2934]
        children = None
      - keys = [2921]
        parent keys = [2916, 2934]
        children = None
      - keys = [2936]
        parent keys = [2916, 2934]
        children = None
  - keys = [3012, 3084]
    parent keys = [2748, 2937]
    children =
    - keys = [2981]
```

```
        parent keys = [3012, 3084]
        children =
        - keys = [2958, 2965]
          parent keys = [2981]
          children = None
        - keys = [2988, 2997]
          parent keys = [2981]
          children = None
      - keys = [3034]
        parent keys = [3012, 3084]
        children =
        - keys = [3013, 3027]
          parent keys = [3034]
          children = None
        - keys = [3040, 3052]
          parent keys = [3034]
          children = None
      - keys = [3140, 3164]
        parent keys = [3012, 3084]
        children =
        - keys = [3088, 3104]
          parent keys = [3140, 3164]
          children = None
        - keys = [3147, 3153]
          parent keys = [3140, 3164]
          children = None
        - keys = [3190]
          parent keys = [3140, 3164]
          children = None
  - keys = [3388, 3544]
    parent keys = [3205, 3725]
    children =
    - keys = [3252, 3311]
      parent keys = [3388, 3544]
      children =
      - keys = [3219]
        parent keys = [3252, 3311]
        children =
        - keys = [3212, 3214]
          parent keys = [3219]
          children = None
        - keys = [3221, 3249]
          parent keys = [3219]
          children = None
      - keys = [3259, 3276]
        parent keys = [3252, 3311]
        children =
        - keys = [3258]
          parent keys = [3259, 3276]
```

```
      children = None
    - keys = [3261]
      parent keys = [3259, 3276]
      children = None
    - keys = [3303]
      parent keys = [3259, 3276]
      children = None
  - keys = [3345, 3361]
    parent keys = [3252, 3311]
    children =
    - keys = [3332, 3338]
      parent keys = [3345, 3361]
      children = None
    - keys = [3348]
      parent keys = [3345, 3361]
      children = None
    - keys = [3385]
      parent keys = [3345, 3361]
      children = None
- keys = [3426, 3478]
  parent keys = [3388, 3544]
  children =
  - keys = [3403]
    parent keys = [3426, 3478]
    children =
    - keys = [3400]
      parent keys = [3403]
      children = None
    - keys = [3406, 3419]
      parent keys = [3403]
      children = None
  - keys = [3454, 3465]
    parent keys = [3426, 3478]
    children =
    - keys = [3452]
      parent keys = [3454, 3465]
      children = None
    - keys = [3464]
      parent keys = [3454, 3465]
      children = None
    - keys = [3472, 3473]
      parent keys = [3454, 3465]
      children = None
  - keys = [3511]
    parent keys = [3426, 3478]
    children =
    - keys = [3482, 3490]
      parent keys = [3511]
      children = None
```

```
          - keys = [3527, 3542]
            parent keys = [3511]
            children = None
      - keys = [3583, 3628]
        parent keys = [3388, 3544]
        children =
        - keys = [3564]
          parent keys = [3583, 3628]
          children =
          - keys = [3559]
            parent keys = [3564]
            children = None
          - keys = [3568]
            parent keys = [3564]
            children = None
        - keys = [3608, 3622]
          parent keys = [3583, 3628]
          children =
          - keys = [3596, 3598]
            parent keys = [3608, 3622]
            children = None
          - keys = [3619]
            parent keys = [3608, 3622]
            children = None
          - keys = [3624]
            parent keys = [3608, 3622]
            children = None
        - keys = [3649, 3706]
          parent keys = [3583, 3628]
          children =
          - keys = [3632, 3646]
            parent keys = [3649, 3706]
            children = None
          - keys = [3660, 3674]
            parent keys = [3649, 3706]
            children = None
          - keys = [3710, 3723]
            parent keys = [3649, 3706]
            children = None
    - keys = [3947, 4133]
      parent keys = [3205, 3725]
      children =
      - keys = [3812, 3934]
        parent keys = [3947, 4133]
        children =
        - keys = [3765, 3787]
          parent keys = [3812, 3934]
          children =
          - keys = [3760]
```

```
      parent keys = [3765, 3787]
      children = None
    - keys = [3772]
      parent keys = [3765, 3787]
      children = None
    - keys = [3790, 3791]
      parent keys = [3765, 3787]
      children = None
  - keys = [3867, 3871]
    parent keys = [3812, 3934]
    children =
    - keys = [3848]
      parent keys = [3867, 3871]
      children = None
    - keys = [3868]
      parent keys = [3867, 3871]
      children = None
    - keys = [3899, 3902]
      parent keys = [3867, 3871]
      children = None
  - keys = [3938]
    parent keys = [3812, 3934]
    children =
    - keys = [3937]
      parent keys = [3938]
      children = None
    - keys = [3942]
      parent keys = [3938]
      children = None
- keys = [4092]
  parent keys = [3947, 4133]
  children =
  - keys = [4047, 4079]
    parent keys = [4092]
    children =
    - keys = [4008, 4023]
      parent keys = [4047, 4079]
      children = None
    - keys = [4071, 4073]
      parent keys = [4047, 4079]
      children = None
    - keys = [4083]
      parent keys = [4047, 4079]
      children = None
  - keys = [4106]
    parent keys = [4092]
    children =
    - keys = [4093]
      parent keys = [4106]
```

```
            children = None
          - keys = [4131]
            parent keys = [4106]
            children = None
      - keys = [4168]
        parent keys = [3947, 4133]
        children =
        - keys = [4155]
          parent keys = [4168]
          children =
          - keys = [4140, 4142]
            parent keys = [4155]
            children = None
          - keys = [4158, 4164]
            parent keys = [4155]
            children = None
        - keys = [4186, 4200]
          parent keys = [4168]
          children =
          - keys = [4170, 4174]
            parent keys = [4186, 4200]
            children = None
          - keys = [4196]
            parent keys = [4186, 4200]
            children = None
          - keys = [4212]
            parent keys = [4186, 4200]
            children = None
- keys = [4845, 5346]
  parent keys = [4269, 5687]
  children =
  - keys = [4470, 4616]
    parent keys = [4845, 5346]
    children =
    - keys = [4327, 4418]
      parent keys = [4470, 4616]
      children =
      - keys = [4286, 4294]
        parent keys = [4327, 4418]
        children =
        - keys = [4272]
          parent keys = [4286, 4294]
          children = None
        - keys = [4291]
          parent keys = [4286, 4294]
          children = None
        - keys = [4302, 4314]
          parent keys = [4286, 4294]
          children = None
```

```
    - keys = [4339, 4374]
      parent keys = [4327, 4418]
      children =
      - keys = [4335]
        parent keys = [4339, 4374]
        children = None
      - keys = [4361]
        parent keys = [4339, 4374]
        children = None
      - keys = [4390, 4399]
        parent keys = [4339, 4374]
        children = None
    - keys = [4432, 4454]
      parent keys = [4327, 4418]
      children =
      - keys = [4425]
        parent keys = [4432, 4454]
        children = None
      - keys = [4439]
        parent keys = [4432, 4454]
        children = None
      - keys = [4456, 4459]
        parent keys = [4432, 4454]
        children = None
  - keys = [4487, 4563]
    parent keys = [4470, 4616]
    children =
    - keys = [4473]
      parent keys = [4487, 4563]
      children =
      - keys = [4472]
        parent keys = [4473]
        children = None
      - keys = [4486]
        parent keys = [4473]
        children = None
    - keys = [4518, 4540]
      parent keys = [4487, 4563]
      children =
      - keys = [4494, 4505]
        parent keys = [4518, 4540]
        children = None
      - keys = [4536]
        parent keys = [4518, 4540]
        children = None
      - keys = [4554, 4560]
        parent keys = [4518, 4540]
        children = None
    - keys = [4574, 4591]
```

```
      parent keys = [4487, 4563]
      children =
      - keys = [4567]
        parent keys = [4574, 4591]
        children = None
      - keys = [4583, 4584]
        parent keys = [4574, 4591]
        children = None
      - keys = [4604, 4615]
        parent keys = [4574, 4591]
        children = None
  - keys = [4706, 4770]
    parent keys = [4470, 4616]
    children =
    - keys = [4646, 4691]
      parent keys = [4706, 4770]
      children =
      - keys = [4621, 4633]
        parent keys = [4646, 4691]
        children = None
      - keys = [4660, 4686]
        parent keys = [4646, 4691]
        children = None
      - keys = [4693, 4702]
        parent keys = [4646, 4691]
        children = None
    - keys = [4718, 4742]
      parent keys = [4706, 4770]
      children =
      - keys = [4708]
        parent keys = [4718, 4742]
        children = None
      - keys = [4729]
        parent keys = [4718, 4742]
        children = None
      - keys = [4763]
        parent keys = [4718, 4742]
        children = None
    - keys = [4805, 4828]
      parent keys = [4706, 4770]
      children =
      - keys = [4772]
        parent keys = [4805, 4828]
        children = None
      - keys = [4811, 4819]
        parent keys = [4805, 4828]
        children = None
      - keys = [4838]
        parent keys = [4805, 4828]
```

```
          children = None
    - keys = [4964, 5189]
      parent keys = [4845, 5346]
      children =
      - keys = [4921]
        parent keys = [4964, 5189]
        children =
        - keys = [4864, 4893]
          parent keys = [4921]
          children =
          - keys = [4849]
            parent keys = [4864, 4893]
            children = None
          - keys = [4865, 4891]
            parent keys = [4864, 4893]
            children = None
          - keys = [4896, 4911]
            parent keys = [4864, 4893]
            children = None
        - keys = [4938, 4959]
          parent keys = [4921]
          children =
          - keys = [4925, 4933]
            parent keys = [4938, 4959]
            children = None
          - keys = [4955, 4957]
            parent keys = [4938, 4959]
            children = None
          - keys = [4960]
            parent keys = [4938, 4959]
            children = None
      - keys = [5041, 5071]
        parent keys = [4964, 5189]
        children =
        - keys = [5018]
          parent keys = [5041, 5071]
          children =
          - keys = [4975, 5002]
            parent keys = [5018]
            children = None
          - keys = [5019, 5039]
            parent keys = [5018]
            children = None
        - keys = [5053]
          parent keys = [5041, 5071]
          children =
          - keys = [5052]
            parent keys = [5053]
            children = None
```

```
            - keys = [5068]
              parent keys = [5053]
              children = None
          - keys = [5092, 5149]
            parent keys = [5041, 5071]
            children =
            - keys = [5078, 5081]
              parent keys = [5092, 5149]
              children = None
            - keys = [5144]
              parent keys = [5092, 5149]
              children = None
            - keys = [5176]
              parent keys = [5092, 5149]
              children = None
        - keys = [5254, 5289]
          parent keys = [4964, 5189]
          children =
          - keys = [5234]
            parent keys = [5254, 5289]
            children =
            - keys = [5190]
              parent keys = [5234]
              children = None
            - keys = [5244, 5250]
              parent keys = [5234]
              children = None
          - keys = [5273]
            parent keys = [5254, 5289]
            children =
            - keys = [5268]
              parent keys = [5273]
              children = None
            - keys = [5275, 5284]
              parent keys = [5273]
              children = None
          - keys = [5310, 5326]
            parent keys = [5254, 5289]
            children =
            - keys = [5301]
              parent keys = [5310, 5326]
              children = None
            - keys = [5319]
              parent keys = [5310, 5326]
              children = None
            - keys = [5331, 5334]
              parent keys = [5310, 5326]
              children = None
      - keys = [5440]
```

```
parent keys = [4845, 5346]
children =
- keys = [5410]
  parent keys = [5440]
  children =
  - keys = [5370]
    parent keys = [5410]
    children =
    - keys = [5352, 5369]
      parent keys = [5370]
      children = None
    - keys = [5376, 5378]
      parent keys = [5370]
      children = None
  - keys = [5434]
    parent keys = [5410]
    children =
    - keys = [5432]
      parent keys = [5434]
      children = None
    - keys = [5435]
      parent keys = [5434]
      children = None
- keys = [5520, 5635]
  parent keys = [5440]
  children =
  - keys = [5488, 5505]
    parent keys = [5520, 5635]
    children =
    - keys = [5485, 5487]
      parent keys = [5488, 5505]
      children = None
    - keys = [5504]
      parent keys = [5488, 5505]
      children = None
    - keys = [5519]
      parent keys = [5488, 5505]
      children = None
  - keys = [5567, 5618]
    parent keys = [5520, 5635]
    children =
    - keys = [5523, 5552]
      parent keys = [5567, 5618]
      children = None
    - keys = [5603]
      parent keys = [5567, 5618]
      children = None
    - keys = [5623]
      parent keys = [5567, 5618]
```

```
            children = None
        - keys = [5664]
          parent keys = [5520, 5635]
          children =
          - keys = [5640]
            parent keys = [5664]
            children = None
          - keys = [5678]
            parent keys = [5664]
            children = None
- keys = [6222, 6409]
  parent keys = [4269, 5687]
  children =
  - keys = [5797, 6017]
    parent keys = [6222, 6409]
    children =
    - keys = [5757]
      parent keys = [5797, 6017]
      children =
      - keys = [5712]
        parent keys = [5757]
        children =
        - keys = [5697, 5698]
          parent keys = [5712]
          children = None
        - keys = [5736, 5737]
          parent keys = [5712]
          children = None
      - keys = [5783]
        parent keys = [5757]
        children =
        - keys = [5770]
          parent keys = [5783]
          children = None
        - keys = [5789, 5795]
          parent keys = [5783]
          children = None
    - keys = [5877, 5960]
      parent keys = [5797, 6017]
      children =
      - keys = [5805, 5819]
        parent keys = [5877, 5960]
        children =
        - keys = [5798, 5800]
          parent keys = [5805, 5819]
          children = None
        - keys = [5814]
          parent keys = [5805, 5819]
          children = None
```

```
      - keys = [5825]
        parent keys = [5805, 5819]
        children = None
    - keys = [5940, 5944]
      parent keys = [5877, 5960]
      children =
      - keys = [5908, 5917]
        parent keys = [5940, 5944]
        children = None
      - keys = [5943]
        parent keys = [5940, 5944]
        children = None
      - keys = [5950]
        parent keys = [5940, 5944]
        children = None
    - keys = [6011]
      parent keys = [5877, 5960]
      children =
      - keys = [6001]
        parent keys = [6011]
        children = None
      - keys = [6013]
        parent keys = [6011]
        children = None
  - keys = [6059, 6143]
    parent keys = [5797, 6017]
    children =
    - keys = [6025, 6039]
      parent keys = [6059, 6143]
      children =
      - keys = [6024]
        parent keys = [6025, 6039]
        children = None
      - keys = [6030, 6036]
        parent keys = [6025, 6039]
        children = None
      - keys = [6046, 6053]
        parent keys = [6025, 6039]
        children = None
    - keys = [6092, 6116]
      parent keys = [6059, 6143]
      children =
      - keys = [6061, 6062]
        parent keys = [6092, 6116]
        children = None
      - keys = [6094, 6113]
        parent keys = [6092, 6116]
        children = None
      - keys = [6136]
```

```
            parent keys = [6092, 6116]
            children = None
        - keys = [6185]
          parent keys = [6059, 6143]
          children =
          - keys = [6158, 6172]
            parent keys = [6185]
            children = None
          - keys = [6201, 6215]
            parent keys = [6185]
            children = None
    - keys = [6303]
      parent keys = [6222, 6409]
      children =
      - keys = [6271]
        parent keys = [6303]
        children =
        - keys = [6235, 6249]
          parent keys = [6271]
          children =
          - keys = [6231]
            parent keys = [6235, 6249]
            children = None
          - keys = [6244]
            parent keys = [6235, 6249]
            children = None
          - keys = [6252, 6263]
            parent keys = [6235, 6249]
            children = None
        - keys = [6275]
          parent keys = [6271]
          children =
          - keys = [6273]
            parent keys = [6275]
            children = None
          - keys = [6300]
            parent keys = [6275]
            children = None
      - keys = [6360]
        parent keys = [6303]
        children =
        - keys = [6331]
          parent keys = [6360]
          children =
          - keys = [6313, 6314]
            parent keys = [6331]
            children = None
          - keys = [6340, 6349]
            parent keys = [6331]
```

```
            children = None
        - keys = [6366, 6394]
          parent keys = [6360]
          children =
          - keys = [6362, 6364]
            parent keys = [6366, 6394]
            children = None
          - keys = [6380, 6393]
            parent keys = [6366, 6394]
            children = None
          - keys = [6398, 6408]
            parent keys = [6366, 6394]
            children = None
    - keys = [6731]
      parent keys = [6222, 6409]
      children =
      - keys = [6528, 6596]
        parent keys = [6731]
        children =
        - keys = [6430, 6477]
          parent keys = [6528, 6596]
          children =
          - keys = [6417]
            parent keys = [6430, 6477]
            children = None
          - keys = [6465]
            parent keys = [6430, 6477]
            children = None
          - keys = [6480, 6527]
            parent keys = [6430, 6477]
            children = None
        - keys = [6553, 6585]
          parent keys = [6528, 6596]
          children =
          - keys = [6546, 6551]
            parent keys = [6553, 6585]
            children = None
          - keys = [6568, 6584]
            parent keys = [6553, 6585]
            children = None
          - keys = [6590]
            parent keys = [6553, 6585]
            children = None
        - keys = [6674, 6682]
          parent keys = [6528, 6596]
          children =
          - keys = [6600, 6603]
            parent keys = [6674, 6682]
            children = None
```

```
          - keys = [6677]
            parent keys = [6674, 6682]
            children = None
          - keys = [6687]
            parent keys = [6674, 6682]
            children = None
      - keys = [6761, 6822]
        parent keys = [6731]
        children =
        - keys = [6733, 6747]
          parent keys = [6761, 6822]
          children =
          - keys = [6732]
            parent keys = [6733, 6747]
            children = None
          - keys = [6734, 6737]
            parent keys = [6733, 6747]
            children = None
          - keys = [6753]
            parent keys = [6733, 6747]
            children = None
        - keys = [6784]
          parent keys = [6761, 6822]
          children =
          - keys = [6775]
            parent keys = [6784]
            children = None
          - keys = [6789, 6802]
            parent keys = [6784]
            children = None
        - keys = [6858, 6906]
          parent keys = [6761, 6822]
          children =
          - keys = [6836, 6840]
            parent keys = [6858, 6906]
            children = None
          - keys = [6886]
            parent keys = [6858, 6906]
            children = None
          - keys = [6918]
            parent keys = [6858, 6906]
            children = None
- keys = [8598, 9400]
  parent keys = [2542, 6923]
  children =
  - keys = [7498, 8179]
    parent keys = [8598, 9400]
    children =
    - keys = [7108, 7261]
```

```
parent keys = [7498, 8179]
children =
- keys = [6996, 7038]
  parent keys = [7108, 7261]
  children =
  - keys = [6935, 6941]
    parent keys = [6996, 7038]
    children =
    - keys = [6924, 6931]
      parent keys = [6935, 6941]
      children = None
    - keys = [6938, 6939]
      parent keys = [6935, 6941]
      children = None
    - keys = [6973, 6986]
      parent keys = [6935, 6941]
      children = None
  - keys = [7025, 7034]
    parent keys = [6996, 7038]
    children =
    - keys = [7006, 7008]
      parent keys = [7025, 7034]
      children = None
    - keys = [7031]
      parent keys = [7025, 7034]
      children = None
    - keys = [7037]
      parent keys = [7025, 7034]
      children = None
  - keys = [7051, 7059]
    parent keys = [6996, 7038]
    children =
    - keys = [7043]
      parent keys = [7051, 7059]
      children = None
    - keys = [7054]
      parent keys = [7051, 7059]
      children = None
    - keys = [7091, 7098]
      parent keys = [7051, 7059]
      children = None
- keys = [7152, 7202]
  parent keys = [7108, 7261]
  children =
  - keys = [7140]
    parent keys = [7152, 7202]
    children =
    - keys = [7117, 7118]
      parent keys = [7140]
```

```
      children = None
    - keys = [7148]
      parent keys = [7140]
      children = None
  - keys = [7162]
    parent keys = [7152, 7202]
    children =
    - keys = [7161]
      parent keys = [7162]
      children = None
    - keys = [7178, 7183]
      parent keys = [7162]
      children = None
  - keys = [7217, 7243]
    parent keys = [7152, 7202]
    children =
    - keys = [7213, 7214]
      parent keys = [7217, 7243]
      children = None
    - keys = [7227]
      parent keys = [7217, 7243]
      children = None
    - keys = [7256]
      parent keys = [7217, 7243]
      children = None
- keys = [7338, 7432]
  parent keys = [7108, 7261]
  children =
  - keys = [7283, 7309]
    parent keys = [7338, 7432]
    children =
    - keys = [7265]
      parent keys = [7283, 7309]
      children = None
    - keys = [7291]
      parent keys = [7283, 7309]
      children = None
    - keys = [7313, 7316]
      parent keys = [7283, 7309]
      children = None
  - keys = [7417]
    parent keys = [7338, 7432]
    children =
    - keys = [7340, 7364]
      parent keys = [7417]
      children = None
    - keys = [7428]
      parent keys = [7417]
      children = None
```

```
      - keys = [7445, 7472]
        parent keys = [7338, 7432]
        children =
        - keys = [7436, 7440]
          parent keys = [7445, 7472]
          children = None
        - keys = [7462, 7469]
          parent keys = [7445, 7472]
          children = None
        - keys = [7477, 7484]
          parent keys = [7445, 7472]
          children = None
    - keys = [7619, 7788]
      parent keys = [7498, 8179]
      children =
      - keys = [7562]
        parent keys = [7619, 7788]
        children =
        - keys = [7533, 7537]
          parent keys = [7562]
          children =
          - keys = [7530]
            parent keys = [7533, 7537]
            children = None
          - keys = [7534]
            parent keys = [7533, 7537]
            children = None
          - keys = [7545, 7546]
            parent keys = [7533, 7537]
            children = None
        - keys = [7593]
          parent keys = [7562]
          children =
          - keys = [7567, 7579]
            parent keys = [7593]
            children = None
          - keys = [7609, 7618]
            parent keys = [7593]
            children = None
      - keys = [7692]
        parent keys = [7619, 7788]
        children =
        - keys = [7665]
          parent keys = [7692]
          children =
          - keys = [7629]
            parent keys = [7665]
            children = None
          - keys = [7683]
```

```
      parent keys = [7665]
      children = None
    - keys = [7760, 7779]
      parent keys = [7692]
      children =
      - keys = [7720, 7722]
        parent keys = [7760, 7779]
        children = None
      - keys = [7770]
        parent keys = [7760, 7779]
        children = None
      - keys = [7783]
        parent keys = [7760, 7779]
        children = None
  - keys = [7885, 8026]
    parent keys = [7619, 7788]
    children =
    - keys = [7800, 7831]
      parent keys = [7885, 8026]
      children =
      - keys = [7795, 7798]
        parent keys = [7800, 7831]
        children = None
      - keys = [7828]
        parent keys = [7800, 7831]
        children = None
      - keys = [7843, 7845]
        parent keys = [7800, 7831]
        children = None
    - keys = [7965, 8003]
      parent keys = [7885, 8026]
      children =
      - keys = [7902, 7937]
        parent keys = [7965, 8003]
        children = None
      - keys = [7971, 8000]
        parent keys = [7965, 8003]
        children = None
      - keys = [8010, 8020]
        parent keys = [7965, 8003]
        children = None
    - keys = [8098, 8150]
      parent keys = [7885, 8026]
      children =
      - keys = [8065, 8072]
        parent keys = [8098, 8150]
        children = None
      - keys = [8111, 8143]
        parent keys = [8098, 8150]
```

```
            children = None
          - keys = [8163, 8177]
            parent keys = [8098, 8150]
            children = None
    - keys = [8308, 8427]
      parent keys = [7498, 8179]
      children =
      - keys = [8206, 8224]
        parent keys = [8308, 8427]
        children =
        - keys = [8189]
          parent keys = [8206, 8224]
          children =
          - keys = [8184, 8186]
            parent keys = [8189]
            children = None
          - keys = [8191, 8203]
            parent keys = [8189]
            children = None
        - keys = [8220]
          parent keys = [8206, 8224]
          children =
          - keys = [8215]
            parent keys = [8220]
            children = None
          - keys = [8221]
            parent keys = [8220]
            children = None
        - keys = [8243, 8263]
          parent keys = [8206, 8224]
          children =
          - keys = [8227, 8232]
            parent keys = [8243, 8263]
            children = None
          - keys = [8250, 8257]
            parent keys = [8243, 8263]
            children = None
          - keys = [8307]
            parent keys = [8243, 8263]
            children = None
      - keys = [8351, 8384]
        parent keys = [8308, 8427]
        children =
        - keys = [8321, 8333]
          parent keys = [8351, 8384]
          children =
          - keys = [8309, 8319]
            parent keys = [8321, 8333]
            children = None
```

```
            - keys = [8322, 8327]
              parent keys = [8321, 8333]
              children = None
            - keys = [8341, 8342]
              parent keys = [8321, 8333]
              children = None
          - keys = [8369]
            parent keys = [8351, 8384]
            children =
            - keys = [8365]
              parent keys = [8369]
              children = None
            - keys = [8375, 8378]
              parent keys = [8369]
              children = None
          - keys = [8409]
[...(truncated)...]rent keys = [1570]
            children =
            - keys = [1575, 1581]
              parent keys = [1592, 1606]
              children = None
            - keys = [1598]
              parent keys = [1592, 1606]
              children = None
            - keys = [1614, 1625]
              parent keys = [1592, 1606]
              children = None
        - keys = [1712]
          parent keys = [1638, 1754]
          children =
          - keys = [1685]
            parent keys = [1712]
            children =
            - keys = [1659, 1681]
              parent keys = [1685]
              children = None
            - keys = [1692, 1694]
              parent keys = [1685]
              children = None
          - keys = [1730]
            parent keys = [1712]
            children =
            - keys = [1718, 1727]
              parent keys = [1730]
              children = None
            - keys = [1752]
              parent keys = [1730]
              children = None
        - keys = [1806, 1862]
```

```
      parent keys = [1638, 1754]
      children =
      - keys = [1780]
        parent keys = [1806, 1862]
        children =
        - keys = [1764, 1771]
          parent keys = [1780]
          children = None
        - keys = [1800]
          parent keys = [1780]
          children = None
      - keys = [1820, 1836]
        parent keys = [1806, 1862]
        children =
        - keys = [1810]
          parent keys = [1820, 1836]
          children = None
        - keys = [1825, 1830]
          parent keys = [1820, 1836]
          children = None
        - keys = [1849, 1861]
          parent keys = [1820, 1836]
          children = None
      - keys = [1867, 1915]
        parent keys = [1806, 1862]
        children =
        - keys = [1866]
          parent keys = [1867, 1915]
          children = None
        - keys = [1892, 1912]
          parent keys = [1867, 1915]
          children = None
        - keys = [1946, 1955]
          parent keys = [1867, 1915]
          children = None
  - keys = [2096, 2304]
    parent keys = [1500, 1959]
    children =
    - keys = [2042]
      parent keys = [2096, 2304]
      children =
      - keys = [2011]
        parent keys = [2042]
        children =
        - keys = [1963, 1990]
          parent keys = [2011]
          children = None
        - keys = [2012, 2028]
          parent keys = [2011]
```

```
      children = None
    - keys = [2060]
      parent keys = [2042]
      children =
      - keys = [2056]
        parent keys = [2060]
        children = None
      - keys = [2071]
        parent keys = [2060]
        children = None
  - keys = [2160, 2253]
    parent keys = [2096, 2304]
    children =
    - keys = [2118, 2144]
      parent keys = [2160, 2253]
      children =
      - keys = [2105]
        parent keys = [2118, 2144]
        children = None
      - keys = [2128, 2134]
        parent keys = [2118, 2144]
        children = None
      - keys = [2151, 2157]
        parent keys = [2118, 2144]
        children = None
    - keys = [2202, 2212]
      parent keys = [2160, 2253]
      children =
      - keys = [2188, 2195]
        parent keys = [2202, 2212]
        children = None
      - keys = [2211]
        parent keys = [2202, 2212]
        children = None
      - keys = [2216, 2232]
        parent keys = [2202, 2212]
        children = None
    - keys = [2273, 2284]
      parent keys = [2160, 2253]
      children =
      - keys = [2259, 2271]
        parent keys = [2273, 2284]
        children = None
      - keys = [2283]
        parent keys = [2273, 2284]
        children = None
      - keys = [2296, 2299]
        parent keys = [2273, 2284]
        children = None
```

```
        - keys = [2355, 2475]
          parent keys = [2096, 2304]
          children =
          - keys = [2315]
            parent keys = [2355, 2475]
            children =
            - keys = [2309]
              parent keys = [2315]
              children = None
            - keys = [2332, 2334]
              parent keys = [2315]
              children = None
          - keys = [2408]
            parent keys = [2355, 2475]
            children =
            - keys = [2384]
              parent keys = [2408]
              children = None
            - keys = [2420]
              parent keys = [2408]
              children = None
          - keys = [2493, 2523]
            parent keys = [2355, 2475]
            children =
            - keys = [2483, 2487]
              parent keys = [2493, 2523]
              children = None
            - keys = [2494]
              parent keys = [2493, 2523]
              children = None
            - keys = [2540]
              parent keys = [2493, 2523]
              children = None
- keys = [4269, 5687]
  parent keys = [2542, 6923]
  children =
  - keys = [3205, 3725]
    parent keys = [4269, 5687]
    children =
    - keys = [2748, 2937]
      parent keys = [3205, 3725]
      children =
      - keys = [2606, 2654]
        parent keys = [2748, 2937]
        children =
        - keys = [2583]
          parent keys = [2606, 2654]
          children =
          - keys = [2576]
```

```
      parent keys = [2583]
      children = None
    - keys = [2600]
      parent keys = [2583]
      children = None
  - keys = [2636]
    parent keys = [2606, 2654]
    children =
    - keys = [2611, 2618]
      parent keys = [2636]
      children = None
    - keys = [2638, 2645]
      parent keys = [2636]
      children = None
  - keys = [2697, 2718]
    parent keys = [2606, 2654]
    children =
    - keys = [2660, 2696]
      parent keys = [2697, 2718]
      children = None
    - keys = [2698, 2714]
      parent keys = [2697, 2718]
      children = None
    - keys = [2734]
      parent keys = [2697, 2718]
      children = None
- keys = [2833, 2880]
  parent keys = [2748, 2937]
  children =
  - keys = [2769]
    parent keys = [2833, 2880]
    children =
    - keys = [2758, 2764]
      parent keys = [2769]
      children = None
    - keys = [2773, 2781]
      parent keys = [2769]
      children = None
  - keys = [2856]
    parent keys = [2833, 2880]
    children =
    - keys = [2846]
      parent keys = [2856]
      children = None
    - keys = [2857]
      parent keys = [2856]
      children = None
  - keys = [2916, 2934]
    parent keys = [2833, 2880]
```

```
      children =
      - keys = [2896, 2897]
        parent keys = [2916, 2934]
        children = None
      - keys = [2921]
        parent keys = [2916, 2934]
        children = None
      - keys = [2936]
        parent keys = [2916, 2934]
        children = None
    - keys = [3012, 3084]
      parent keys = [2748, 2937]
      children =
      - keys = [2981]
        parent keys = [3012, 3084]
        children =
        - keys = [2958, 2965]
          parent keys = [2981]
          children = None
        - keys = [2988, 2997]
          parent keys = [2981]
          children = None
      - keys = [3034]
        parent keys = [3012, 3084]
        children =
        - keys = [3013, 3027]
          parent keys = [3034]
          children = None
        - keys = [3040, 3052]
          parent keys = [3034]
          children = None
      - keys = [3140, 3164]
        parent keys = [3012, 3084]
        children =
        - keys = [3088, 3104]
          parent keys = [3140, 3164]
          children = None
        - keys = [3147, 3153]
          parent keys = [3140, 3164]
          children = None
        - keys = [3190]
          parent keys = [3140, 3164]
          children = None
  - keys = [3388, 3544]
    parent keys = [3205, 3725]
    children =
    - keys = [3252, 3311]
      parent keys = [3388, 3544]
      children =
```

```
      - keys = [3219]
        parent keys = [3252, 3311]
        children =
        - keys = [3212, 3214]
          parent keys = [3219]
          children = None
        - keys = [3221, 3249]
          parent keys = [3219]
          children = None
      - keys = [3259, 3276]
        parent keys = [3252, 3311]
        children =
        - keys = [3258]
          parent keys = [3259, 3276]
          children = None
        - keys = [3261]
          parent keys = [3259, 3276]
          children = None
        - keys = [3303]
          parent keys = [3259, 3276]
          children = None
      - keys = [3345, 3361]
        parent keys = [3252, 3311]
        children =
        - keys = [3332, 3338]
          parent keys = [3345, 3361]
          children = None
        - keys = [3348]
          parent keys = [3345, 3361]
          children = None
        - keys = [3385]
          parent keys = [3345, 3361]
          children = None
    - keys = [3426, 3478]
      parent keys = [3388, 3544]
      children =
      - keys = [3403]
        parent keys = [3426, 3478]
        children =
        - keys = [3400]
          parent keys = [3403]
          children = None
        - keys = [3406, 3419]
          parent keys = [3403]
          children = None
      - keys = [3454, 3465]
        parent keys = [3426, 3478]
        children =
        - keys = [3452]
```

```
      parent keys = [3454, 3465]
      children = None
    - keys = [3464]
      parent keys = [3454, 3465]
      children = None
    - keys = [3472, 3473]
      parent keys = [3454, 3465]
      children = None
  - keys = [3511]
    parent keys = [3426, 3478]
    children =
    - keys = [3482, 3490]
      parent keys = [3511]
      children = None
    - keys = [3527, 3542]
      parent keys = [3511]
      children = None
- keys = [3583, 3628]
  parent keys = [3388, 3544]
  children =
  - keys = [3564]
    parent keys = [3583, 3628]
    children =
    - keys = [3559]
      parent keys = [3564]
      children = None
    - keys = [3568]
      parent keys = [3564]
      children = None
  - keys = [3608, 3622]
    parent keys = [3583, 3628]
    children =
    - keys = [3596, 3598]
      parent keys = [3608, 3622]
      children = None
    - keys = [3619]
      parent keys = [3608, 3622]
      children = None
    - keys = [3624]
      parent keys = [3608, 3622]
      children = None
  - keys = [3649, 3706]
    parent keys = [3583, 3628]
    children =
    - keys = [3632, 3646]
      parent keys = [3649, 3706]
      children = None
    - keys = [3660, 3674]
      parent keys = [3649, 3706]
```

```
          children = None
        - keys = [3710, 3723]
          parent keys = [3649, 3706]
          children = None
- keys = [3947, 4133]
  parent keys = [3205, 3725]
  children =
  - keys = [3812, 3934]
    parent keys = [3947, 4133]
    children =
    - keys = [3765, 3787]
      parent keys = [3812, 3934]
      children =
      - keys = [3760]
        parent keys = [3765, 3787]
        children = None
      - keys = [3772]
        parent keys = [3765, 3787]
        children = None
      - keys = [3790, 3791]
        parent keys = [3765, 3787]
        children = None
    - keys = [3867, 3871]
      parent keys = [3812, 3934]
      children =
      - keys = [3848]
        parent keys = [3867, 3871]
        children = None
      - keys = [3868]
        parent keys = [3867, 3871]
        children = None
      - keys = [3899, 3902]
        parent keys = [3867, 3871]
        children = None
    - keys = [3938]
      parent keys = [3812, 3934]
      children =
      - keys = [3937]
        parent keys = [3938]
        children = None
      - keys = [3942]
        parent keys = [3938]
        children = None
  - keys = [4092]
    parent keys = [3947, 4133]
    children =
    - keys = [4047, 4079]
      parent keys = [4092]
      children =
```

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          - keys = [4008, 4023]
            parent keys = [4047, 4079]
            children = None
          - keys = [4071, 4073]
            parent keys = [4047, 4079]
            children = None
          - keys = [4083]
            parent keys = [4047, 4079]
            children = None
        - keys = [4106]
          parent keys = [4092]
          children =
          - keys = [4093]
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          - keys = [4131]
            parent keys = [4106]
            children = None
      - keys = [4168]
        parent keys = [3947, 4133]
        children =
        - keys = [4155]
          parent keys = [4168]
          children =
          - keys = [4140, 4142]
            parent keys = [4155]
            children = None
          - keys = [4158, 4164]
            parent keys = [4155]
            children = None
        - keys = [4186, 4200]
          parent keys = [4168]
          children =
          - keys = [4170, 4174]
            parent keys = [4186, 4200]
            children = None
          - keys = [4196]
            parent keys = [4186, 4200]
            children = None
          - keys = [4212]
            parent keys = [4186, 4200]
            children = None
  - keys = [4845, 5346]
    parent keys = [4269, 5687]
    children =
    - keys = [4470, 4616]
      parent keys = [4845, 5346]
      children =
      - keys = [4327, 4418]
```

```
    parent keys = [4470, 4616]
    children =
    - keys = [4286, 4294]
      parent keys = [4327, 4418]
      children =
      - keys = [4272]
        parent keys = [4286, 4294]
        children = None
      - keys = [4291]
        parent keys = [4286, 4294]
        children = None
      - keys = [4302, 4314]
        parent keys = [4286, 4294]
        children = None
    - keys = [4339, 4374]
      parent keys = [4327, 4418]
      children =
      - keys = [4335]
        parent keys = [4339, 4374]
        children = None
      - keys = [4361]
        parent keys = [4339, 4374]
        children = None
      - keys = [4390, 4399]
        parent keys = [4339, 4374]
        children = None
    - keys = [4432, 4454]
      parent keys = [4327, 4418]
      children =
      - keys = [4425]
        parent keys = [4432, 4454]
        children = None
      - keys = [4439]
        parent keys = [4432, 4454]
        children = None
      - keys = [4456, 4459]
        parent keys = [4432, 4454]
        children = None
  - keys = [4487, 4563]
    parent keys = [4470, 4616]
    children =
    - keys = [4473]
      parent keys = [4487, 4563]
      children =
      - keys = [4472]
        parent keys = [4473]
        children = None
      - keys = [4486]
        parent keys = [4473]
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      children = None
    - keys = [4518, 4540]
      parent keys = [4487, 4563]
      children =
      - keys = [4494, 4505]
        parent keys = [4518, 4540]
        children = None
      - keys = [4536]
        parent keys = [4518, 4540]
        children = None
      - keys = [4554, 4560]
        parent keys = [4518, 4540]
        children = None
    - keys = [4574, 4591]
      parent keys = [4487, 4563]
      children =
      - keys = [4567]
        parent keys = [4574, 4591]
        children = None
      - keys = [4583, 4584]
        parent keys = [4574, 4591]
        children = None
      - keys = [4604, 4615]
        parent keys = [4574, 4591]
        children = None
  - keys = [4706, 4770]
    parent keys = [4470, 4616]
    children =
    - keys = [4646, 4691]
      parent keys = [4706, 4770]
      children =
      - keys = [4621, 4633]
        parent keys = [4646, 4691]
        children = None
      - keys = [4660, 4686]
        parent keys = [4646, 4691]
        children = None
      - keys = [4693, 4702]
        parent keys = [4646, 4691]
        children = None
    - keys = [4718, 4742]
      parent keys = [4706, 4770]
      children =
      - keys = [4708]
        parent keys = [4718, 4742]
        children = None
      - keys = [4729]
        parent keys = [4718, 4742]
        children = None
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        - keys = [4763]
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          children = None
      - keys = [4805, 4828]
        parent keys = [4706, 4770]
        children =
        - keys = [4772]
          parent keys = [4805, 4828]
          children = None
        - keys = [4811, 4819]
          parent keys = [4805, 4828]
          children = None
        - keys = [4838]
          parent keys = [4805, 4828]
          children = None
- keys = [4964, 5189]
  parent keys = [4845, 5346]
  children =
  - keys = [4921]
    parent keys = [4964, 5189]
    children =
    - keys = [4864, 4893]
      parent keys = [4921]
      children =
      - keys = [4849]
        parent keys = [4864, 4893]
        children = None
      - keys = [4865, 4891]
        parent keys = [4864, 4893]
        children = None
      - keys = [4896, 4911]
        parent keys = [4864, 4893]
        children = None
    - keys = [4938, 4959]
      parent keys = [4921]
      children =
      - keys = [4925, 4933]
        parent keys = [4938, 4959]
        children = None
      - keys = [4955, 4957]
        parent keys = [4938, 4959]
        children = None
      - keys = [4960]
        parent keys = [4938, 4959]
        children = None
  - keys = [5041, 5071]
    parent keys = [4964, 5189]
    children =
    - keys = [5018]
```

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      children =
      - keys = [4975, 5002]
        parent keys = [5018]
        children = None
      - keys = [5019, 5039]
        parent keys = [5018]
        children = None
    - keys = [5053]
      parent keys = [5041, 5071]
      children =
      - keys = [5052]
        parent keys = [5053]
        children = None
      - keys = [5068]
        parent keys = [5053]
        children = None
    - keys = [5092, 5149]
      parent keys = [5041, 5071]
      children =
      - keys = [5078, 5081]
        parent keys = [5092, 5149]
        children = None
      - keys = [5144]
        parent keys = [5092, 5149]
        children = None
      - keys = [5176]
        parent keys = [5092, 5149]
        children = None
  - keys = [5254, 5289]
    parent keys = [4964, 5189]
    children =
    - keys = [5234]
      parent keys = [5254, 5289]
      children =
      - keys = [5190]
        parent keys = [5234]
        children = None
      - keys = [5244, 5250]
        parent keys = [5234]
        children = None
    - keys = [5273]
      parent keys = [5254, 5289]
      children =
      - keys = [5268]
        parent keys = [5273]
        children = None
      - keys = [5275, 5284]
        parent keys = [5273]
```

```
            children = None
          - keys = [5310, 5326]
            parent keys = [5254, 5289]
            children =
            - keys = [5301]
              parent keys = [5310, 5326]
              children = None
            - keys = [5319]
              parent keys = [5310, 5326]
              children = None
            - keys = [5331, 5334]
              parent keys = [5310, 5326]
              children = None
      - keys = [5440]
        parent keys = [4845, 5346]
        children =
        - keys = [5410]
          parent keys = [5440]
          children =
          - keys = [5370]
            parent keys = [5410]
            children =
            - keys = [5352, 5369]
              parent keys = [5370]
              children = None
            - keys = [5376, 5378]
              parent keys = [5370]
              children = None
          - keys = [5434]
            parent keys = [5410]
            children =
            - keys = [5432]
              parent keys = [5434]
              children = None
            - keys = [5435]
              parent keys = [5434]
              children = None
        - keys = [5520, 5635]
          parent keys = [5440]
          children =
          - keys = [5488, 5505]
            parent keys = [5520, 5635]
            children =
            - keys = [5485, 5487]
              parent keys = [5488, 5505]
              children = None
            - keys = [5504]
              parent keys = [5488, 5505]
              children = None
```

```
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            parent keys = [5488, 5505]
            children = None
        - keys = [5567, 5618]
          parent keys = [5520, 5635]
          children =
          - keys = [5523, 5552]
            parent keys = [5567, 5618]
            children = None
          - keys = [5603]
            parent keys = [5567, 5618]
            children = None
          - keys = [5623]
            parent keys = [5567, 5618]
            children = None
        - keys = [5664]
          parent keys = [5520, 5635]
          children =
          - keys = [5640]
            parent keys = [5664]
            children = None
          - keys = [5678]
            parent keys = [5664]
            children = None
- keys = [6222, 6409]
  parent keys = [4269, 5687]
  children =
  - keys = [5797, 6017]
    parent keys = [6222, 6409]
    children =
    - keys = [5757]
      parent keys = [5797, 6017]
      children =
      - keys = [5712]
        parent keys = [5757]
        children =
        - keys = [5697, 5698]
          parent keys = [5712]
          children = None
        - keys = [5736, 5737]
          parent keys = [5712]
          children = None
      - keys = [5783]
        parent keys = [5757]
        children =
        - keys = [5770]
          parent keys = [5783]
          children = None
        - keys = [5789, 5795]
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      children = None
- keys = [5877, 5960]
  parent keys = [5797, 6017]
  children =
  - keys = [5805, 5819]
    parent keys = [5877, 5960]
    children =
    - keys = [5798, 5800]
      parent keys = [5805, 5819]
      children = None
    - keys = [5814]
      parent keys = [5805, 5819]
      children = None
    - keys = [5825]
      parent keys = [5805, 5819]
      children = None
  - keys = [5940, 5944]
    parent keys = [5877, 5960]
    children =
    - keys = [5908, 5917]
      parent keys = [5940, 5944]
      children = None
    - keys = [5943]
      parent keys = [5940, 5944]
      children = None
    - keys = [5950]
      parent keys = [5940, 5944]
      children = None
  - keys = [6011]
    parent keys = [5877, 5960]
    children =
    - keys = [6001]
      parent keys = [6011]
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    - keys = [6013]
      parent keys = [6011]
      children = None
- keys = [6059, 6143]
  parent keys = [5797, 6017]
  children =
  - keys = [6025, 6039]
    parent keys = [6059, 6143]
    children =
    - keys = [6024]
      parent keys = [6025, 6039]
      children = None
    - keys = [6030, 6036]
      parent keys = [6025, 6039]
```

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            children = None
          - keys = [6046, 6053]
            parent keys = [6025, 6039]
            children = None
        - keys = [6092, 6116]
          parent keys = [6059, 6143]
          children =
          - keys = [6061, 6062]
            parent keys = [6092, 6116]
            children = None
          - keys = [6094, 6113]
            parent keys = [6092, 6116]
            children = None
          - keys = [6136]
            parent keys = [6092, 6116]
            children = None
        - keys = [6185]
          parent keys = [6059, 6143]
          children =
          - keys = [6158, 6172]
            parent keys = [6185]
            children = None
          - keys = [6201, 6215]
            parent keys = [6185]
            children = None
    - keys = [6303]
      parent keys = [6222, 6409]
      children =
      - keys = [6271]
        parent keys = [6303]
        children =
        - keys = [6235, 6249]
          parent keys = [6271]
          children =
          - keys = [6231]
            parent keys = [6235, 6249]
            children = None
          - keys = [6244]
            parent keys = [6235, 6249]
            children = None
          - keys = [6252, 6263]
            parent keys = [6235, 6249]
            children = None
        - keys = [6275]
          parent keys = [6271]
          children =
          - keys = [6273]
            parent keys = [6275]
            children = None
```

```
          - keys = [6300]
            parent keys = [6275]
            children = None
      - keys = [6360]
        parent keys = [6303]
        children =
        - keys = [6331]
          parent keys = [6360]
          children =
          - keys = [6313, 6314]
            parent keys = [6331]
            children = None
          - keys = [6340, 6349]
            parent keys = [6331]
            children = None
        - keys = [6366, 6394]
          parent keys = [6360]
          children =
          - keys = [6362, 6364]
            parent keys = [6366, 6394]
            children = None
          - keys = [6380, 6393]
            parent keys = [6366, 6394]
            children = None
          - keys = [6398, 6408]
            parent keys = [6366, 6394]
            children = None
    - keys = [6731]
      parent keys = [6222, 6409]
      children =
      - keys = [6528, 6596]
        parent keys = [6731]
        children =
        - keys = [6430, 6477]
          parent keys = [6528, 6596]
          children =
          - keys = [6417]
            parent keys = [6430, 6477]
            children = None
          - keys = [6465]
            parent keys = [6430, 6477]
            children = None
          - keys = [6480, 6527]
            parent keys = [6430, 6477]
            children = None
        - keys = [6553, 6585]
          parent keys = [6528, 6596]
          children =
          - keys = [6546, 6551]
```

```
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      children = None
    - keys = [6568, 6584]
      parent keys = [6553, 6585]
      children = None
    - keys = [6590]
      parent keys = [6553, 6585]
      children = None
  - keys = [6674, 6682]
    parent keys = [6528, 6596]
    children =
    - keys = [6600, 6603]
      parent keys = [6674, 6682]
      children = None
    - keys = [6677]
      parent keys = [6674, 6682]
      children = None
    - keys = [6687]
      parent keys = [6674, 6682]
      children = None
- keys = [6761, 6822]
  parent keys = [6731]
  children =
  - keys = [6733, 6747]
    parent keys = [6761, 6822]
    children =
    - keys = [6732]
      parent keys = [6733, 6747]
      children = None
    - keys = [6734, 6737]
      parent keys = [6733, 6747]
      children = None
    - keys = [6753]
      parent keys = [6733, 6747]
      children = None
  - keys = [6784]
    parent keys = [6761, 6822]
    children =
    - keys = [6775]
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    - keys = [6789, 6802]
      parent keys = [6784]
      children = None
  - keys = [6858, 6906]
    parent keys = [6761, 6822]
    children =
    - keys = [6836, 6840]
      parent keys = [6858, 6906]
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```
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            children = None
          - keys = [6918]
            parent keys = [6858, 6906]
            children = None
- keys = [8598, 9400]
  parent keys = [2542, 6923]
  children =
  - keys = [7498, 8179]
    parent keys = [8598, 9400]
    children =
    - keys = [7108, 7261]
      parent keys = [7498, 8179]
      children =
      - keys = [6996, 7038]
        parent keys = [7108, 7261]
        children =
        - keys = [6935, 6941]
          parent keys = [6996, 7038]
          children =
          - keys = [6924, 6931]
            parent keys = [6935, 6941]
            children = None
          - keys = [6938, 6939]
            parent keys = [6935, 6941]
            children = None
          - keys = [6973, 6986]
            parent keys = [6935, 6941]
            children = None
        - keys = [7025, 7034]
          parent keys = [6996, 7038]
          children =
          - keys = [7006, 7008]
            parent keys = [7025, 7034]
            children = None
          - keys = [7031]
            parent keys = [7025, 7034]
            children = None
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            children = None
        - keys = [7051, 7059]
          parent keys = [6996, 7038]
          children =
          - keys = [7043]
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            children = None
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      - keys = [7091, 7098]
        parent keys = [7051, 7059]
        children = None
  - keys = [7152, 7202]
    parent keys = [7108, 7261]
    children =
    - keys = [7140]
      parent keys = [7152, 7202]
      children =
      - keys = [7117, 7118]
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      - keys = [7148]
        parent keys = [7140]
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    - keys = [7162]
      parent keys = [7152, 7202]
      children =
      - keys = [7161]
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      - keys = [7178, 7183]
        parent keys = [7162]
        children = None
    - keys = [7217, 7243]
      parent keys = [7152, 7202]
      children =
      - keys = [7213, 7214]
        parent keys = [7217, 7243]
        children = None
      - keys = [7227]
        parent keys = [7217, 7243]
        children = None
      - keys = [7256]
        parent keys = [7217, 7243]
        children = None
  - keys = [7338, 7432]
    parent keys = [7108, 7261]
    children =
    - keys = [7283, 7309]
      parent keys = [7338, 7432]
      children =
      - keys = [7265]
        parent keys = [7283, 7309]
        children = None
      - keys = [7291]
```

```
            parent keys = [7283, 7309]
            children = None
          - keys = [7313, 7316]
            parent keys = [7283, 7309]
            children = None
        - keys = [7417]
          parent keys = [7338, 7432]
          children =
          - keys = [7340, 7364]
            parent keys = [7417]
            children = None
          - keys = [7428]
            parent keys = [7417]
            children = None
        - keys = [7445, 7472]
          parent keys = [7338, 7432]
          children =
          - keys = [7436, 7440]
            parent keys = [7445, 7472]
            children = None
          - keys = [7462, 7469]
            parent keys = [7445, 7472]
            children = None
          - keys = [7477, 7484]
            parent keys = [7445, 7472]
            children = None
    - keys = [7619, 7788]
      parent keys = [7498, 8179]
      children =
      - keys = [7562]
        parent keys = [7619, 7788]
        children =
        - keys = [7533, 7537]
          parent keys = [7562]
          children =
          - keys = [7530]
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            children = None
          - keys = [7534]
            parent keys = [7533, 7537]
            children = None
          - keys = [7545, 7546]
            parent keys = [7533, 7537]
            children = None
        - keys = [7593]
          parent keys = [7562]
          children =
          - keys = [7567, 7579]
            parent keys = [7593]
```

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      parent keys = [7593]
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- keys = [7692]
  parent keys = [7619, 7788]
  children =
  - keys = [7665]
    parent keys = [7692]
    children =
    - keys = [7629]
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      children = None
    - keys = [7683]
      parent keys = [7665]
      children = None
  - keys = [7760, 7779]
    parent keys = [7692]
    children =
    - keys = [7720, 7722]
      parent keys = [7760, 7779]
      children = None
    - keys = [7770]
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      children = None
    - keys = [7783]
      parent keys = [7760, 7779]
      children = None
- keys = [7885, 8026]
  parent keys = [7619, 7788]
  children =
  - keys = [7800, 7831]
    parent keys = [7885, 8026]
    children =
    - keys = [7795, 7798]
      parent keys = [7800, 7831]
      children = None
    - keys = [7828]
      parent keys = [7800, 7831]
      children = None
    - keys = [7843, 7845]
      parent keys = [7800, 7831]
      children = None
  - keys = [7965, 8003]
    parent keys = [7885, 8026]
    children =
    - keys = [7902, 7937]
      parent keys = [7965, 8003]
      children = None
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          - keys = [8010, 8020]
            parent keys = [7965, 8003]
            children = None
        - keys = [8098, 8150]
          parent keys = [7885, 8026]
          children =
          - keys = [8065, 8072]
            parent keys = [8098, 8150]
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          - keys = [8111, 8143]
            parent keys = [8098, 8150]
            children = None
          - keys = [8163, 8177]
            parent keys = [8098, 8150]
            children = None
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      parent keys = [7498, 8179]
      children =
      - keys = [8206, 8224]
        parent keys = [8308, 8427]
        children =
        - keys = [8189]
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          - keys = [8184, 8186]
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        - keys = [8220]
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          - keys = [8215]
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          - keys = [8221]
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            children = None
        - keys = [8243, 8263]
          parent keys = [8206, 8224]
          children =
          - keys = [8227, 8232]
            parent keys = [8243, 8263]
            children = None
          - keys = [8250, 8257]
```

```
      parent keys = [8243, 8263]
      children = None
    - keys = [8307]
      parent keys = [8243, 8263]
      children = None
- keys = [8351, 8384]
  parent keys = [8308, 8427]
  children =
  - keys = [8321, 8333]
    parent keys = [8351, 8384]
    children =
    - keys = [8309, 8319]
      parent keys = [8321, 8333]
      children = None
    - keys = [8322, 8327]
      parent keys = [8321, 8333]
      children = None
    - keys = [8341, 8342]
      parent keys = [8321, 8333]
      children = None
  - keys = [8369]
    parent keys = [8351, 8384]
    children =
    - keys = [8365]
      parent keys = [8369]
      children = None
    - keys = [8375, 8378]
      parent keys = [8369]
      children = None
  - keys = [8409]
    parent keys = [8351, 8384]
    children =
    - keys = [8403, 8404]
      parent keys = [8409]
      children = None
    - keys = [8410, 8419]
      parent keys = [8409]
      children = None
- keys = [8485]
  parent keys = [8308, 8427]
  children =
  - keys = [8439]
    parent keys = [8485]
    children =
    - keys = [8438]
      parent keys = [8439]
      children = None
    - keys = [8475, 8483]
      parent keys = [8439]
```

```
            children = None
        - keys = [8564]
          parent keys = [8485]
          children =
          - keys = [8486, 8525]
            parent keys = [8564]
            children = None
          - keys = [8574, 8583]
            parent keys = [8564]
            children = None
- keys = [8879]
  parent keys = [8598, 9400]
  children =
  - keys = [8778]
    parent keys = [8879]
    children =
    - keys = [8689]
      parent keys = [8778]
      children =
      - keys = [8607, 8637]
        parent keys = [8689]
        children =
        - keys = [8601]
          parent keys = [8607, 8637]
          children = None
        - keys = [8613]
          parent keys = [8607, 8637]
          children = None
        - keys = [8642, 8671]
          parent keys = [8607, 8637]
          children = None
      - keys = [8699, 8719]
        parent keys = [8689]
        children =
        - keys = [8695, 8698]
          parent keys = [8699, 8719]
          children = None
        - keys = [8717, 8718]
          parent keys = [8699, 8719]
          children = None
        - keys = [8738, 8754]
          parent keys = [8699, 8719]
          children = None
    - keys = [8806]
      parent keys = [8778]
      children =
      - keys = [8797]
        parent keys = [8806]
        children =
```

```
          - keys = [8787, 8796]
            parent keys = [8797]
            children = None
          - keys = [8801]
            parent keys = [8797]
            children = None
        - keys = [8844, 8866]
          parent keys = [8806]
          children =
          - keys = [8817, 8822]
            parent keys = [8844, 8866]
            children = None
          - keys = [8858]
            parent keys = [8844, 8866]
            children = None
          - keys = [8871]
            parent keys = [8844, 8866]
            children = None
    - keys = [9056, 9185]
      parent keys = [8879]
      children =
      - keys = [8927, 8967]
        parent keys = [9056, 9185]
        children =
        - keys = [8893, 8915]
          parent keys = [8927, 8967]
          children =
          - keys = [8889]
            parent keys = [8893, 8915]
            children = None
          - keys = [8894, 8896]
            parent keys = [8893, 8915]
            children = None
          - keys = [8916, 8918]
            parent keys = [8893, 8915]
            children = None
        - keys = [8948]
          parent keys = [8927, 8967]
          children =
          - keys = [8942, 8945]
            parent keys = [8948]
            children = None
          - keys = [8955]
            parent keys = [8948]
            children = None
        - keys = [9010, 9031]
          parent keys = [8927, 8967]
          children =
          - keys = [9004, 9009]
```

```
      parent keys = [9010, 9031]
      children = None
    - keys = [9026, 9030]
      parent keys = [9010, 9031]
      children = None
    - keys = [9034, 9053]
      parent keys = [9010, 9031]
      children = None
- keys = [9121, 9138]
  parent keys = [9056, 9185]
  children =
  - keys = [9099, 9111]
    parent keys = [9121, 9138]
    children =
    - keys = [9070, 9075]
      parent keys = [9099, 9111]
      children = None
    - keys = [9100, 9102]
      parent keys = [9099, 9111]
      children = None
    - keys = [9114]
      parent keys = [9099, 9111]
      children = None
  - keys = [9125]
    parent keys = [9121, 9138]
    children =
    - keys = [9123]
      parent keys = [9125]
      children = None
    - keys = [9129, 9133]
      parent keys = [9125]
      children = None
  - keys = [9146, 9180]
    parent keys = [9121, 9138]
    children =
    - keys = [9139, 9142]
      parent keys = [9146, 9180]
      children = None
    - keys = [9149, 9176]
      parent keys = [9146, 9180]
      children = None
    - keys = [9183]
      parent keys = [9146, 9180]
      children = None
- keys = [9287]
  parent keys = [9056, 9185]
  children =
  - keys = [9196, 9215]
    parent keys = [9287]
```

```
          children =
          - keys = [9188, 9195]
            parent keys = [9196, 9215]
            children = None
          - keys = [9202, 9206]
            parent keys = [9196, 9215]
            children = None
          - keys = [9218, 9265]
            parent keys = [9196, 9215]
            children = None
        - keys = [9309, 9369]
          parent keys = [9287]
          children =
          - keys = [9302, 9305]
            parent keys = [9309, 9369]
            children = None
          - keys = [9352, 9359]
            parent keys = [9309, 9369]
            children = None
          - keys = [9373, 9390]
            parent keys = [9309, 9369]
            children = None
- keys = [9758]
  parent keys = [8598, 9400]
  children =
  - keys = [9495, 9578]
    parent keys = [9758]
    children =
    - keys = [9427]
      parent keys = [9495, 9578]
      children =
      - keys = [9407, 9419]
        parent keys = [9427]
        children =
        - keys = [9406]
          parent keys = [9407, 9419]
          children = None
        - keys = [9409, 9412]
          parent keys = [9407, 9419]
          children = None
        - keys = [9420, 9426]
          parent keys = [9407, 9419]
          children = None
      - keys = [9468]
        parent keys = [9427]
        children =
        - keys = [9442, 9449]
          parent keys = [9468]
          children = None
```

```
      - keys = [9481, 9482]
        parent keys = [9468]
        children = None
  - keys = [9535]
    parent keys = [9495, 9578]
    children =
    - keys = [9509]
      parent keys = [9535]
      children =
      - keys = [9504]
        parent keys = [9509]
        children = None
      - keys = [9517, 9532]
        parent keys = [9509]
        children = None
    - keys = [9553, 9569]
      parent keys = [9535]
      children =
      - keys = [9539, 9542]
        parent keys = [9553, 9569]
        children = None
      - keys = [9554, 9559]
        parent keys = [9553, 9569]
        children = None
      - keys = [9575, 9576]
        parent keys = [9553, 9569]
        children = None
  - keys = [9636, 9703]
    parent keys = [9495, 9578]
    children =
    - keys = [9590, 9619]
      parent keys = [9636, 9703]
      children =
      - keys = [9586, 9588]
        parent keys = [9590, 9619]
        children = None
      - keys = [9614, 9617]
        parent keys = [9590, 9619]
        children = None
      - keys = [9624, 9629]
        parent keys = [9590, 9619]
        children = None
    - keys = [9683]
      parent keys = [9636, 9703]
      children =
      - keys = [9644]
        parent keys = [9683]
        children = None
      - keys = [9691]
```

```
            parent keys = [9683]
            children = None
        - keys = [9742]
          parent keys = [9636, 9703]
          children =
          - keys = [9717, 9725]
            parent keys = [9742]
            children = None
          - keys = [9743, 9744]
            parent keys = [9742]
            children = None
    - keys = [9875]
      parent keys = [9758]
      children =
      - keys = [9791]
        parent keys = [9875]
        children =
        - keys = [9771, 9777]
          parent keys = [9791]
          children =
          - keys = [9763]
            parent keys = [9771, 9777]
            children = None
          - keys = [9774]
            parent keys = [9771, 9777]
            children = None
          - keys = [9786]
            parent keys = [9771, 9777]
            children = None
        - keys = [9825, 9848]
          parent keys = [9791]
          children =
          - keys = [9802]
            parent keys = [9825, 9848]
            children = None
          - keys = [9837, 9840]
            parent keys = [9825, 9848]
            children = None
          - keys = [9851, 9866]
            parent keys = [9825, 9848]
            children = None
      - keys = [9929]
        parent keys = [9875]
        children =
        - keys = [9892]
          parent keys = [9929]
          children =
          - keys = [9876, 9887]
            parent keys = [9892]
```

```
              children = None
            - keys = [9916, 9919]
              parent keys = [9892]
              children = None
          - keys = [9946, 9972]
            parent keys = [9929]
            children =
            - keys = [9943]
              parent keys = [9946, 9972]
              children = None
            - keys = [9948]
              parent keys = [9946, 9972]
              children = None
            - keys = [9983, 9986]
              parent keys = [9946, 9972]
              children = None

Sub-Tests #2-#5: Passed (Detailed Omitted)
```

8) test_8 (test_generated_cases.simplebtree) (0/10)

```
Test Set 8: Performing 5 tests with size = 1000 and type = insert and d
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,6
insert,9320
insert,7016
.
.
.
insert,6272
insert,200
dump
---> Your Solution:
- keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [139, 688]
    parent keys = [1880, 4793, 7438, 9048]
    children =
    - keys = [53, 100]
      parent keys = [139, 688]
      children =
      - keys = [21, 29, 36, 38, 40]
        parent keys = [53, 100]
        children = None
      - keys = [84, 89, 92, 93]
        parent keys = [53, 100]
        children = None
      - keys = [112, 116, 121, 124, 128]
        parent keys = [53, 100]
        children = None
    - keys = [184, 330, 495]
      parent keys = [139, 688]
      children =
      - keys = [145, 148, 154, 163, 183]
        parent keys = [184, 330, 495]
        children = None
      - keys = [200, 273]
        parent keys = [184, 330, 495]
        children = None
      - keys = [351, 359, 393]
        parent keys = [184, 330, 495]
        children = None
      - keys = [579, 614, 617, 642, 660]
        parent keys = [184, 330, 495]
        children = None
    - keys = [961, 1024, 1206, 1471, 1687]
      parent keys = [139, 688]
```

```
      children =
      - keys = [718, 728, 830, 897, 921]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
      - keys = [979, 1016]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
      - keys = [1046, 1063, 1102]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
      - keys = [1284, 1342, 1374, 1413, 1445]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
      - keys = [1499, 1559, 1564, 1606, 1675]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
      - keys = [1731, 1776, 1817, 1856]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
  - keys = [3083, 3898, 4223]
    parent keys = [1880, 4793, 7438, 9048]
    children =
    - keys = [2179, 2574, 2723]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [1893, 1914, 1925, 1946, 2023]
        parent keys = [2179, 2574, 2723]
        children = None
      - keys = [2295, 2389, 2457, 2483, 2546]
        parent keys = [2179, 2574, 2723]
        children = None
      - keys = [2602, 2646, 2665, 2716]
        parent keys = [2179, 2574, 2723]
        children = None
      - keys = [2772, 2809, 2879, 2916, 3076]
        parent keys = [2179, 2574, 2723]
        children = None
    - keys = [3217, 3448, 3618, 3671]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [3143, 3156, 3193, 3201, 3213]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3254, 3278, 3294, 3331]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3490, 3495, 3509, 3538, 3556]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
```

```
      - keys = [3633, 3663]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3768, 3863, 3878]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
    - keys = [3954, 4000, 4081, 4123, 4172]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [3900, 3909, 3912, 3914, 3943]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [3959, 3960, 3974, 3992]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [4002, 4007, 4033, 4045]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [4086, 4095, 4105, 4107, 4121]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [4127, 4146, 4152, 4161, 4166]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [4207, 4213, 4218]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
    - keys = [4253, 4438, 4515, 4688, 4771]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [4224, 4234, 4235, 4245, 4246]
        parent keys = [4253, 4438, 4515, 4688, 4771]
        children = None
      - keys = [4257, 4268, 4277, 4330]
        parent keys = [4253, 4438, 4515, 4688, 4771]
        children = None
      - keys = [4447, 4487, 4504]
        parent keys = [4253, 4438, 4515, 4688, 4771]
        children = None
      - keys = [4597, 4616, 4624, 4646, 4678]
        parent keys = [4253, 4438, 4515, 4688, 4771]
        children = None
      - keys = [4710, 4735]
        parent keys = [4253, 4438, 4515, 4688, 4771]
        children = None
      - keys = [4772, 4774, 4781]
        parent keys = [4253, 4438, 4515, 4688, 4771]
        children = None
  - keys = [4997, 5318, 6033, 6532, 6995]
```

```
parent keys = [1880, 4793, 7438, 9048]
children =
- keys = [4799, 4891]
  parent keys = [4997, 5318, 6033, 6532, 6995]
  children =
  - keys = [4794, 4797]
    parent keys = [4799, 4891]
    children = None
  - keys = [4825, 4840, 4841, 4869, 4879]
    parent keys = [4799, 4891]
    children = None
  - keys = [4892, 4907, 4934, 4937, 4970]
    parent keys = [4799, 4891]
    children = None
- keys = [5078, 5133, 5221]
  parent keys = [4997, 5318, 6033, 6532, 6995]
  children =
  - keys = [4999, 5007, 5026, 5057, 5061]
    parent keys = [5078, 5133, 5221]
    children = None
  - keys = [5088, 5101]
    parent keys = [5078, 5133, 5221]
    children = None
  - keys = [5137, 5151, 5159, 5201, 5205]
    parent keys = [5078, 5133, 5221]
    children = None
  - keys = [5225, 5263, 5265, 5294, 5308]
    parent keys = [5078, 5133, 5221]
    children = None
- keys = [5414, 5453, 5599, 5799, 5937]
  parent keys = [4997, 5318, 6033, 6532, 6995]
  children =
  - keys = [5327, 5335, 5366, 5407, 5411]
    parent keys = [5414, 5453, 5599, 5799, 5937]
    children = None
  - keys = [5420, 5446]
    parent keys = [5414, 5453, 5599, 5799, 5937]
    children = None
  - keys = [5483, 5490, 5501, 5530, 5556]
    parent keys = [5414, 5453, 5599, 5799, 5937]
    children = None
  - keys = [5611, 5661, 5718, 5758, 5768]
    parent keys = [5414, 5453, 5599, 5799, 5937]
    children = None
  - keys = [5809, 5815, 5823, 5865, 5932]
    parent keys = [5414, 5453, 5599, 5799, 5937]
    children = None
  - keys = [5955, 5990, 6004, 6032]
    parent keys = [5414, 5453, 5599, 5799, 5937]
```

```
      children = None
  - keys = [6132, 6197, 6272, 6372, 6506]
    parent keys = [4997, 5318, 6033, 6532, 6995]
    children =
    - keys = [6037, 6057, 6059, 6087, 6094]
      parent keys = [6132, 6197, 6272, 6372, 6506]
      children = None
    - keys = [6137, 6158, 6174, 6185, 6193]
      parent keys = [6132, 6197, 6272, 6372, 6506]
      children = None
    - keys = [6211, 6236, 6255, 6260, 6264]
      parent keys = [6132, 6197, 6272, 6372, 6506]
      children = None
    - keys = [6276, 6293, 6333, 6352, 6353]
      parent keys = [6132, 6197, 6272, 6372, 6506]
      children = None
    - keys = [6387, 6446, 6474, 6484, 6492]
      parent keys = [6132, 6197, 6272, 6372, 6506]
      children = None
    - keys = [6510, 6522]
      parent keys = [6132, 6197, 6272, 6372, 6506]
      children = None
  - keys = [6584, 6710, 6781, 6935]
    parent keys = [4997, 5318, 6033, 6532, 6995]
    children =
    - keys = [6574, 6581, 6583]
      parent keys = [6584, 6710, 6781, 6935]
      children = None
    - keys = [6610, 6621, 6634, 6637, 6677]
      parent keys = [6584, 6710, 6781, 6935]
      children = None
    - keys = [6749, 6763, 6765, 6767, 6778]
      parent keys = [6584, 6710, 6781, 6935]
      children = None
    - keys = [6807, 6831, 6865, 6899]
      parent keys = [6584, 6710, 6781, 6935]
      children = None
    - keys = [6947, 6949, 6975]
      parent keys = [6584, 6710, 6781, 6935]
      children = None
  - keys = [7021, 7237, 7386]
    parent keys = [4997, 5318, 6033, 6532, 6995]
    children =
    - keys = [6996, 6998, 7014]
      parent keys = [7021, 7237, 7386]
      children = None
    - keys = [7042, 7061, 7142, 7164, 7194]
      parent keys = [7021, 7237, 7386]
      children = None
```

```
    - keys = [7340, 7346, 7373, 7382]
      parent keys = [7021, 7237, 7386]
      children = None
    - keys = [7412, 7415, 7430, 7434]
      parent keys = [7021, 7237, 7386]
      children = None
- keys = [7815, 8360, 8751]
  parent keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [7514, 7627, 7719, 7748]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [7445, 7455, 7478, 7484, 7485]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7530, 7550, 7568, 7583, 7590]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7643, 7657, 7686, 7702, 7703]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7727, 7731, 7732, 7734]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7769, 7770, 7774, 7788, 7791]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
  - keys = [7903, 7999, 8120, 8175, 8288]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [7827, 7888]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [7938, 7972, 7976, 7998]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8041, 8058, 8076, 8077, 8096]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8136, 8157]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8225, 8227, 8242]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8312, 8332, 8338, 8339, 8344]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
  - keys = [8496, 8550]
```

```
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [8481, 8490]
      parent keys = [8496, 8550]
      children = None
    - keys = [8506, 8539, 8549]
      parent keys = [8496, 8550]
      children = None
    - keys = [8567, 8570, 8623, 8665, 8742]
      parent keys = [8496, 8550]
      children = None
  - keys = [8770, 8890, 8995]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [8760, 8765]
      parent keys = [8770, 8890, 8995]
      children = None
    - keys = [8783, 8791, 8862, 8865, 8875]
      parent keys = [8770, 8890, 8995]
      children = None
    - keys = [8899, 8928, 8978, 8989, 8990]
      parent keys = [8770, 8890, 8995]
      children = None
    - keys = [8999, 9004, 9014, 9040]
      parent keys = [8770, 8890, 8995]
      children = None
- keys = [9430, 9752, 9837]
  parent keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [9102, 9176, 9234, 9300]
    parent keys = [9430, 9752, 9837]
    children =
    - keys = [9056, 9064, 9078, 9088, 9096]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9116, 9123, 9145, 9157, 9171]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9209, 9220, 9221]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9255, 9266, 9270, 9283, 9285]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9320, 9362, 9389, 9401]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
  - keys = [9458, 9504, 9578, 9622, 9675]
    parent keys = [9430, 9752, 9837]
```

```
      children =
      - keys = [9435, 9437, 9438, 9440, 9447]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
      - keys = [9460, 9471, 9473, 9480, 9494]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
      - keys = [9531, 9536, 9544, 9555, 9570]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
      - keys = [9590, 9594, 9604]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
      - keys = [9630, 9633, 9654, 9672]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
      - keys = [9676, 9693, 9713, 9717, 9727]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
    - keys = [9776, 9823]
      parent keys = [9430, 9752, 9837]
      children =
      - keys = [9755, 9774]
        parent keys = [9776, 9823]
        children = None
      - keys = [9783, 9794, 9801, 9804, 9813]
        parent keys = [9776, 9823]
        children = None
      - keys = [9831, 9833]
        parent keys = [9776, 9823]
        children = None
    - keys = [9892, 9936, 9964]
      parent keys = [9430, 9752, 9837]
      children =
      - keys = [9852, 9857, 9891]
        parent keys = [9892, 9936, 9964]
        children = None
      - keys = [9893, 9900, 9915, 9916, 9920]
        parent keys = [9892, 9936, 9964]
        children = None
      - keys = [9937, 9948]
        parent keys = [9892, 9936, 9964]
        children = None
      - keys = [9971, 9983, 9990, 9998]
        parent keys = [9892, 9936, 9964]
        children = None

---> Correct Solution:
- keys = [1880, 4793, 7438, 9048]
```

```
children =
- keys = [139, 688]
  parent keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [53, 100]
    parent keys = [139, 688]
    children =
    - keys = [21, 29, 36, 38, 40]
      parent keys = [53, 100]
      children = None
    - keys = [84, 89, 92, 93]
      parent keys = [53, 100]
      children = None
    - keys = [112, 116, 121, 124, 128]
      parent keys = [53, 100]
      children = None
  - keys = [184, 330, 495]
    parent keys = [139, 688]
    children =
    - keys = [145, 148, 154, 163, 183]
      parent keys = [184, 330, 495]
      children = None
    - keys = [200, 273]
      parent keys = [184, 330, 495]
      children = None
    - keys = [351, 359, 393]
      parent keys = [184, 330, 495]
      children = None
    - keys = [579, 614, 617, 642, 660]
      parent keys = [184, 330, 495]
      children = None
  - keys = [961, 1024, 1206, 1471, 1687]
    parent keys = [139, 688]
    children =
    - keys = [718, 728, 830, 897, 921]
      parent keys = [961, 1024, 1206, 1471, 1687]
      children = None
    - keys = [979, 1016]
      parent keys = [961, 1024, 1206, 1471, 1687]
      children = None
    - keys = [1046, 1063, 1102]
      parent keys = [961, 1024, 1206, 1471, 1687]
      children = None
    - keys = [1284, 1342, 1374, 1413, 1445]
      parent keys = [961, 1024, 1206, 1471, 1687]
      children = None
    - keys = [1499, 1559, 1564, 1606, 1675]
      parent keys = [961, 1024, 1206, 1471, 1687]
      children = None
```

```
      - keys = [1731, 1776, 1817, 1856]
        parent keys = [961, 1024, 1206, 1471, 1687]
        children = None
  - keys = [3083, 3898, 4223]
    parent keys = [1880, 4793, 7438, 9048]
    children =
    - keys = [2179, 2574, 2723]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [1893, 1914, 1925, 1946, 2023]
        parent keys = [2179, 2574, 2723]
        children = None
      - keys = [2295, 2389, 2457, 2483, 2546]
        parent keys = [2179, 2574, 2723]
        children = None
      - keys = [2602, 2646, 2665, 2716]
        parent keys = [2179, 2574, 2723]
        children = None
      - keys = [2772, 2809, 2879, 2916, 3076]
        parent keys = [2179, 2574, 2723]
        children = None
    - keys = [3217, 3448, 3618, 3671]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [3143, 3156, 3193, 3201, 3213]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3254, 3278, 3294, 3331]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3490, 3495, 3509, 3538, 3556]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3633, 3663]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
      - keys = [3768, 3863, 3878]
        parent keys = [3217, 3448, 3618, 3671]
        children = None
    - keys = [3954, 4000, 4081, 4123, 4172]
      parent keys = [3083, 3898, 4223]
      children =
      - keys = [3900, 3909, 3912, 3914, 3943]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [3959, 3960, 3974, 3992]
        parent keys = [3954, 4000, 4081, 4123, 4172]
        children = None
      - keys = [4002, 4007, 4033, 4045]
```

```
      parent keys = [3954, 4000, 4081, 4123, 4172]
      children = None
    - keys = [4086, 4095, 4105, 4107, 4121]
      parent keys = [3954, 4000, 4081, 4123, 4172]
      children = None
    - keys = [4127, 4146, 4152, 4161, 4166]
      parent keys = [3954, 4000, 4081, 4123, 4172]
      children = None
    - keys = [4207, 4213, 4218]
      parent keys = [3954, 4000, 4081, 4123, 4172]
      children = None
  - keys = [4253, 4438, 4515, 4688, 4771]
    parent keys = [3083, 3898, 4223]
    children =
    - keys = [4224, 4234, 4235, 4245, 4246]
      parent keys = [4253, 4438, 4515, 4688, 4771]
      children = None
    - keys = [4257, 4268, 4277, 4330]
      parent keys = [4253, 4438, 4515, 4688, 4771]
      children = None
    - keys = [4447, 4487, 4504]
      parent keys = [4253, 4438, 4515, 4688, 4771]
      children = None
    - keys = [4597, 4616, 4624, 4646, 4678]
      parent keys = [4253, 4438, 4515, 4688, 4771]
      children = None
    - keys = [4710, 4735]
      parent keys = [4253, 4438, 4515, 4688, 4771]
      children = None
    - keys = [4772, 4774, 4781]
      parent keys = [4253, 4438, 4515, 4688, 4771]
      children = None
- keys = [4997, 5318, 6033, 6532, 6995]
  parent keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [4799, 4891]
    parent keys = [4997, 5318, 6033, 6532, 6995]
    children =
    - keys = [4794, 4797]
      parent keys = [4799, 4891]
      children = None
    - keys = [4825, 4840, 4841, 4869, 4879]
      parent keys = [4799, 4891]
      children = None
    - keys = [4892, 4907, 4934, 4937, 4970]
      parent keys = [4799, 4891]
      children = None
  - keys = [5078, 5133, 5221]
    parent keys = [4997, 5318, 6033, 6532, 6995]
```

```
      children =
      - keys = [4999, 5007, 5026, 5057, 5061]
        parent keys = [5078, 5133, 5221]
        children = None
      - keys = [5088, 5101]
        parent keys = [5078, 5133, 5221]
        children = None
      - keys = [5137, 5151, 5159, 5201, 5205]
        parent keys = [5078, 5133, 5221]
        children = None
      - keys = [5225, 5263, 5265, 5294, 5308]
        parent keys = [5078, 5133, 5221]
        children = None
    - keys = [5414, 5453, 5599, 5799, 5937]
      parent keys = [4997, 5318, 6033, 6532, 6995]
      children =
      - keys = [5327, 5335, 5366, 5407, 5411]
        parent keys = [5414, 5453, 5599, 5799, 5937]
        children = None
      - keys = [5420, 5446]
        parent keys = [5414, 5453, 5599, 5799, 5937]
        children = None
      - keys = [5483, 5490, 5501, 5530, 5556]
        parent keys = [5414, 5453, 5599, 5799, 5937]
        children = None
      - keys = [5611, 5661, 5718, 5758, 5768]
        parent keys = [5414, 5453, 5599, 5799, 5937]
        children = None
      - keys = [5809, 5815, 5823, 5865, 5932]
        parent keys = [5414, 5453, 5599, 5799, 5937]
        children = None
      - keys = [5955, 5990, 6004, 6032]
        parent keys = [5414, 5453, 5599, 5799, 5937]
        children = None
    - keys = [6132, 6197, 6272, 6372, 6506]
      parent keys = [4997, 5318, 6033, 6532, 6995]
      children =
      - keys = [6037, 6057, 6059, 6087, 6094]
        parent keys = [6132, 6197, 6272, 6372, 6506]
        children = None
      - keys = [6137, 6158, 6174, 6185, 6193]
        parent keys = [6132, 6197, 6272, 6372, 6506]
        children = None
      - keys = [6211, 6236, 6255, 6260, 6264]
        parent keys = [6132, 6197, 6272, 6372, 6506]
        children = None
      - keys = [6276, 6293, 6333, 6352, 6353]
        parent keys = [6132, 6197, 6272, 6372, 6506]
        children = None
```

```
      - keys = [6387, 6446, 6474, 6484, 6492]
        parent keys = [6132, 6197, 6272, 6372, 6506]
        children = None
      - keys = [6510, 6522]
        parent keys = [6132, 6197, 6272, 6372, 6506]
        children = None
    - keys = [6584, 6710, 6781, 6935]
      parent keys = [4997, 5318, 6033, 6532, 6995]
      children =
      - keys = [6574, 6581, 6583]
        parent keys = [6584, 6710, 6781, 6935]
        children = None
      - keys = [6610, 6621, 6634, 6637, 6677]
        parent keys = [6584, 6710, 6781, 6935]
        children = None
      - keys = [6749, 6763, 6765, 6767, 6778]
        parent keys = [6584, 6710, 6781, 6935]
        children = None
      - keys = [6807, 6831, 6865, 6899]
        parent keys = [6584, 6710, 6781, 6935]
        children = None
      - keys = [6947, 6949, 6975]
        parent keys = [6584, 6710, 6781, 6935]
        children = None
    - keys = [7021, 7237, 7386]
      parent keys = [4997, 5318, 6033, 6532, 6995]
      children =
      - keys = [6996, 6998, 7014]
        parent keys = [7021, 7237, 7386]
        children = None
      - keys = [7042, 7061, 7142, 7164, 7194]
        parent keys = [7021, 7237, 7386]
        children = None
      - keys = [7340, 7346, 7373, 7382]
        parent keys = [7021, 7237, 7386]
        children = None
      - keys = [7412, 7415, 7430, 7434]
        parent keys = [7021, 7237, 7386]
        children = None
- keys = [7815, 8360, 8751]
  parent keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [7514, 7627, 7719, 7748]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [7445, 7455, 7478, 7484, 7485]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7530, 7550, 7568, 7583, 7590]
```

```
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7643, 7657, 7686, 7702, 7703]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7727, 7731, 7732, 7734]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
    - keys = [7769, 7770, 7774, 7788, 7791]
      parent keys = [7514, 7627, 7719, 7748]
      children = None
  - keys = [7903, 7999, 8120, 8175, 8288]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [7827, 7888]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [7938, 7972, 7976, 7998]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8041, 8058, 8076, 8077, 8096]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8136, 8157]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8225, 8227, 8242]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
    - keys = [8312, 8332, 8338, 8339, 8344]
      parent keys = [7903, 7999, 8120, 8175, 8288]
      children = None
  - keys = [8496, 8550]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [8481, 8490]
      parent keys = [8496, 8550]
      children = None
    - keys = [8506, 8539, 8549]
      parent keys = [8496, 8550]
      children = None
    - keys = [8567, 8570, 8623, 8665, 8742]
      parent keys = [8496, 8550]
      children = None
  - keys = [8770, 8890, 8995]
    parent keys = [7815, 8360, 8751]
    children =
    - keys = [8760, 8765]
      parent keys = [8770, 8890, 8995]
```

```
      children = None
    - keys = [8783, 8791, 8862, 8865, 8875]
      parent keys = [8770, 8890, 8995]
      children = None
    - keys = [8899, 8928, 8978, 8989, 8990]
      parent keys = [8770, 8890, 8995]
      children = None
    - keys = [8999, 9004, 9014, 9040]
      parent keys = [8770, 8890, 8995]
      children = None
- keys = [9430, 9752, 9837]
  parent keys = [1880, 4793, 7438, 9048]
  children =
  - keys = [9102, 9176, 9234, 9300]
    parent keys = [9430, 9752, 9837]
    children =
    - keys = [9056, 9064, 9078, 9088, 9096]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9116, 9123, 9145, 9157, 9171]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9209, 9220, 9221]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9255, 9266, 9270, 9283, 9285]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
    - keys = [9320, 9362, 9389, 9401]
      parent keys = [9102, 9176, 9234, 9300]
      children = None
  - keys = [9458, 9504, 9578, 9622, 9675]
    parent keys = [9430, 9752, 9837]
    children =
    - keys = [9435, 9437, 9438, 9440, 9447]
      parent keys = [9458, 9504, 9578, 9622, 9675]
      children = None
    - keys = [9460, 9471, 9473, 9480, 9494]
      parent keys = [9458, 9504, 9578, 9622, 9675]
      children = None
    - keys = [9531, 9536, 9544, 9555, 9570]
      parent keys = [9458, 9504, 9578, 9622, 9675]
      children = None
    - keys = [9590, 9594, 9604]
      parent keys = [9458, 9504, 9578, 9622, 9675]
      children = None
    - keys = [9630, 9633, 9654, 9672]
      parent keys = [9458, 9504, 9578, 9622, 9675]
      children = None
```

```
      - keys = [9676, 9693, 9713, 9717, 9727]
        parent keys = [9458, 9504, 9578, 9622, 9675]
        children = None
    - keys = [9776, 9823]
      parent keys = [9430, 9752, 9837]
      children =
      - keys = [9755, 9774]
        parent keys = [9776, 9823]
        children = None
      - keys = [9783, 9794, 9801, 9804, 9813]
        parent keys = [9776, 9823]
        children = None
      - keys = [9831, 9833]
        parent keys = [9776, 9823]
        children = None
    - keys = [9892, 9936, 9964]
      parent keys = [9430, 9752, 9837]
      children =
      - keys = [9852, 9857, 9891]
        parent keys = [9892, 9936, 9964]
        children = None
      - keys = [9893, 9900, 9915, 9916, 9920]
        parent keys = [9892, 9936, 9964]
        children = None
      - keys = [9937, 9948]
        parent keys = [9892, 9936, 9964]
        children = None
      - keys = [9971, 9983, 9990, 9998]
        parent keys = [9892, 9936, 9964]
        children = None

Sub-Test #2: Failed (Details follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,6
insert,4755
insert,1751
.
.
.
insert,3418
insert,7539
dump
---> Your Solution:
- keys = [1683, 5303, 7608]
  children =
  - keys = [255, 631, 793, 1074, 1153]
    parent keys = [1683, 5303, 7608]
    children =
```

```
- keys = [33, 79, 121, 183]
  parent keys = [255, 631, 793, 1074, 1153]
  children =
  - keys = [22, 25, 26, 31, 32]
    parent keys = [33, 79, 121, 183]
    children = None
  - keys = [40, 62]
    parent keys = [33, 79, 121, 183]
    children = None
  - keys = [81, 86, 113]
    parent keys = [33, 79, 121, 183]
    children = None
  - keys = [123, 127, 148, 173, 177]
    parent keys = [33, 79, 121, 183]
    children = None
  - keys = [192, 210, 226, 239, 251]
    parent keys = [33, 79, 121, 183]
    children = None
- keys = [302, 404, 475, 504, 553]
  parent keys = [255, 631, 793, 1074, 1153]
  children =
  - keys = [267, 273, 277, 283, 300]
    parent keys = [302, 404, 475, 504, 553]
    children = None
  - keys = [318, 344, 358, 366]
    parent keys = [302, 404, 475, 504, 553]
    children = None
  - keys = [409, 421, 440, 458, 470]
    parent keys = [302, 404, 475, 504, 553]
    children = None
  - keys = [481, 484]
    parent keys = [302, 404, 475, 504, 553]
    children = None
  - keys = [518, 528, 532, 551]
    parent keys = [302, 404, 475, 504, 553]
    children = None
  - keys = [586, 594, 600, 619, 623]
    parent keys = [302, 404, 475, 504, 553]
    children = None
- keys = [662, 686, 724, 728, 765]
  parent keys = [255, 631, 793, 1074, 1153]
  children =
  - keys = [635, 636, 641, 650]
    parent keys = [662, 686, 724, 728, 765]
    children = None
  - keys = [672, 680, 681, 683, 684]
    parent keys = [662, 686, 724, 728, 765]
    children = None
  - keys = [690, 694, 708, 719, 723]
```

```
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [725, 727]
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [736, 749, 757]
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [767, 768, 770, 776, 777]
      parent keys = [662, 686, 724, 728, 765]
      children = None
  - keys = [823, 863, 1007, 1029]
    parent keys = [255, 631, 793, 1074, 1153]
    children =
    - keys = [800, 813]
      parent keys = [823, 863, 1007, 1029]
      children = None
    - keys = [824, 826, 857]
      parent keys = [823, 863, 1007, 1029]
      children = None
    - keys = [873, 900, 912, 958, 993]
      parent keys = [823, 863, 1007, 1029]
      children = None
    - keys = [1019, 1020]
      parent keys = [823, 863, 1007, 1029]
      children = None
    - keys = [1053, 1064, 1071]
      parent keys = [823, 863, 1007, 1029]
      children = None
  - keys = [1091, 1123]
    parent keys = [255, 631, 793, 1074, 1153]
    children =
    - keys = [1075, 1087, 1090]
      parent keys = [1091, 1123]
      children = None
    - keys = [1099, 1104, 1109, 1115]
      parent keys = [1091, 1123]
      children = None
    - keys = [1126, 1127, 1133, 1138, 1152]
      parent keys = [1091, 1123]
      children = None
  - keys = [1214, 1299, 1485]
    parent keys = [255, 631, 793, 1074, 1153]
    children =
    - keys = [1173, 1192, 1199, 1207, 1212]
      parent keys = [1214, 1299, 1485]
      children = None
    - keys = [1222, 1283]
      parent keys = [1214, 1299, 1485]
```

```
      children = None
    - keys = [1337, 1367, 1452, 1479]
      parent keys = [1214, 1299, 1485]
      children = None
    - keys = [1489, 1508, 1523, 1532, 1669]
      parent keys = [1214, 1299, 1485]
      children = None
- keys = [2520, 4262]
  parent keys = [1683, 5303, 7608]
  children =
  - keys = [1862, 1984, 2150, 2383]
    parent keys = [2520, 4262]
    children =
    - keys = [1757, 1777]
      parent keys = [1862, 1984, 2150, 2383]
      children = None
    - keys = [1889, 1909, 1960]
      parent keys = [1862, 1984, 2150, 2383]
      children = None
    - keys = [2031, 2046, 2050, 2098, 2111]
      parent keys = [1862, 1984, 2150, 2383]
      children = None
    - keys = [2210, 2323]
      parent keys = [1862, 1984, 2150, 2383]
      children = None
    - keys = [2410, 2462, 2465, 2483]
      parent keys = [1862, 1984, 2150, 2383]
      children = None
  - keys = [2601, 2767, 3055, 3545, 4041]
    parent keys = [2520, 4262]
    children =
    - keys = [2588, 2591]
      parent keys = [2601, 2767, 3055, 3545, 4041]
      children = None
    - keys = [2625, 2714, 2751]
      parent keys = [2601, 2767, 3055, 3545, 4041]
      children = None
    - keys = [2833, 2906, 2957, 3035, 3040]
      parent keys = [2601, 2767, 3055, 3545, 4041]
      children = None
    - keys = [3075, 3177, 3351, 3418, 3486]
      parent keys = [2601, 2767, 3055, 3545, 4041]
      children = None
    - keys = [3548, 3645, 3793, 3920, 3964]
      parent keys = [2601, 2767, 3055, 3545, 4041]
      children = None
    - keys = [4056, 4142, 4143, 4201]
      parent keys = [2601, 2767, 3055, 3545, 4041]
      children = None
```

```
  - keys = [4392, 4656, 4854]
    parent keys = [2520, 4262]
    children =
    - keys = [4284, 4296, 4336, 4365]
      parent keys = [4392, 4656, 4854]
      children = None
    - keys = [4406, 4456, 4498, 4526, 4532]
      parent keys = [4392, 4656, 4854]
      children = None
    - keys = [4675, 4684, 4737, 4745, 4790]
      parent keys = [4392, 4656, 4854]
      children = None
    - keys = [5015, 5031, 5200, 5220, 5244]
      parent keys = [4392, 4656, 4854]
      children = None
- keys = [5711, 6277, 6579, 7074]
  parent keys = [1683, 5303, 7608]
  children =
  - keys = [5455, 5549, 5610]
    parent keys = [5711, 6277, 6579, 7074]
    children =
    - keys = [5313, 5326]
      parent keys = [5455, 5549, 5610]
      children = None
    - keys = [5471, 5492, 5517]
      parent keys = [5455, 5549, 5610]
      children = None
    - keys = [5551, 5557, 5558, 5567, 5609]
      parent keys = [5455, 5549, 5610]
      children = None
    - keys = [5615, 5631, 5636, 5708]
      parent keys = [5455, 5549, 5610]
      children = None
  - keys = [5858, 5996, 6025, 6113, 6213]
    parent keys = [5711, 6277, 6579, 7074]
    children =
    - keys = [5713, 5766, 5773, 5827, 5837]
      parent keys = [5858, 5996, 6025, 6113, 6213]
      children = None
    - keys = [5899, 5946, 5949, 5974]
      parent keys = [5858, 5996, 6025, 6113, 6213]
      children = None
    - keys = [6003, 6007, 6024]
      parent keys = [5858, 5996, 6025, 6113, 6213]
      children = None
    - keys = [6040, 6061, 6067, 6102, 6104]
      parent keys = [5858, 5996, 6025, 6113, 6213]
      children = None
    - keys = [6173, 6181, 6189]
```

```
      parent keys = [5858, 5996, 6025, 6113, 6213]
      children = None
    - keys = [6217, 6224, 6239]
      parent keys = [5858, 5996, 6025, 6113, 6213]
      children = None
  - keys = [6342, 6401]
    parent keys = [5711, 6277, 6579, 7074]
    children =
    - keys = [6297, 6320, 6337]
      parent keys = [6342, 6401]
      children = None
    - keys = [6377, 6383, 6384, 6392]
      parent keys = [6342, 6401]
      children = None
    - keys = [6409, 6421, 6427, 6522, 6541]
      parent keys = [6342, 6401]
      children = None
  - keys = [6676, 6856, 6877, 7000]
    parent keys = [5711, 6277, 6579, 7074]
    children =
    - keys = [6595, 6604, 6643]
      parent keys = [6676, 6856, 6877, 7000]
      children = None
    - keys = [6704, 6711, 6776, 6816, 6854]
      parent keys = [6676, 6856, 6877, 7000]
      children = None
    - keys = [6865, 6869, 6876]
      parent keys = [6676, 6856, 6877, 7000]
      children = None
    - keys = [6904, 6920, 6982]
      parent keys = [6676, 6856, 6877, 7000]
      children = None
    - keys = [7032, 7043, 7046, 7070, 7073]
      parent keys = [6676, 6856, 6877, 7000]
      children = None
  - keys = [7125, 7221, 7348, 7512, 7539]
    parent keys = [5711, 6277, 6579, 7074]
    children =
    - keys = [7090, 7095]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7127, 7180, 7210, 7213]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7228, 7251, 7252, 7304, 7331]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7371, 7373, 7394, 7437, 7443]
      parent keys = [7125, 7221, 7348, 7512, 7539]
```

```
      children = None
    - keys = [7514, 7536]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7541, 7596, 7600]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
- keys = [8031, 8452, 9003, 9453, 9748]
  parent keys = [1683, 5303, 7608]
  children =
  - keys = [7651, 7741, 7807, 7934, 7956]
    parent keys = [8031, 8452, 9003, 9453, 9748]
    children =
    - keys = [7616, 7617, 7622]
      parent keys = [7651, 7741, 7807, 7934, 7956]
      children = None
    - keys = [7681, 7690, 7698, 7700, 7707]
      parent keys = [7651, 7741, 7807, 7934, 7956]
      children = None
    - keys = [7749, 7756, 7765]
      parent keys = [7651, 7741, 7807, 7934, 7956]
      children = None
    - keys = [7836, 7839, 7840, 7851, 7924]
      parent keys = [7651, 7741, 7807, 7934, 7956]
      children = None
    - keys = [7942, 7947]
      parent keys = [7651, 7741, 7807, 7934, 7956]
      children = None
    - keys = [7980, 7989, 8021]
      parent keys = [7651, 7741, 7807, 7934, 7956]
      children = None
  - keys = [8105, 8204, 8277, 8314, 8340]
    parent keys = [8031, 8452, 9003, 9453, 9748]
    children =
    - keys = [8039, 8040, 8086, 8088, 8091]
      parent keys = [8105, 8204, 8277, 8314, 8340]
      children = None
    - keys = [8106, 8145, 8179, 8189, 8190]
      parent keys = [8105, 8204, 8277, 8314, 8340]
      children = None
    - keys = [8228, 8244, 8271]
      parent keys = [8105, 8204, 8277, 8314, 8340]
      children = None
    - keys = [8282, 8291, 8293, 8301, 8304]
      parent keys = [8105, 8204, 8277, 8314, 8340]
      children = None
    - keys = [8316, 8323]
      parent keys = [8105, 8204, 8277, 8314, 8340]
      children = None
```

```
    - keys = [8351, 8388, 8391, 8419]
      parent keys = [8105, 8204, 8277, 8314, 8340]
      children = None
  - keys = [8536, 8646, 8682, 8777, 8885]
    parent keys = [8031, 8452, 9003, 9453, 9748]
    children =
    - keys = [8473, 8474, 8475, 8513, 8528]
      parent keys = [8536, 8646, 8682, 8777, 8885]
      children = None
    - keys = [8614, 8620]
      parent keys = [8536, 8646, 8682, 8777, 8885]
      children = None
    - keys = [8655, 8657, 8677, 8678]
      parent keys = [8536, 8646, 8682, 8777, 8885]
      children = None
    - keys = [8704, 8713, 8725, 8747, 8773]
      parent keys = [8536, 8646, 8682, 8777, 8885]
      children = None
    - keys = [8791, 8833, 8860]
      parent keys = [8536, 8646, 8682, 8777, 8885]
      children = None
    - keys = [8921, 8986, 8998]
      parent keys = [8536, 8646, 8682, 8777, 8885]
      children = None
  - keys = [9151, 9254, 9289, 9378, 9429]
    parent keys = [8031, 8452, 9003, 9453, 9748]
    children =
    - keys = [9012, 9039, 9074]
      parent keys = [9151, 9254, 9289, 9378, 9429]
      children = None
    - keys = [9163, 9169, 9173, 9180]
      parent keys = [9151, 9254, 9289, 9378, 9429]
      children = None
    - keys = [9260, 9261, 9267, 9270, 9282]
      parent keys = [9151, 9254, 9289, 9378, 9429]
      children = None
    - keys = [9310, 9313, 9314, 9353, 9368]
      parent keys = [9151, 9254, 9289, 9378, 9429]
      children = None
    - keys = [9397, 9404, 9406, 9411, 9418]
      parent keys = [9151, 9254, 9289, 9378, 9429]
      children = None
    - keys = [9431, 9437, 9445, 9448]
      parent keys = [9151, 9254, 9289, 9378, 9429]
      children = None
  - keys = [9483, 9544, 9607, 9673, 9710]
    parent keys = [8031, 8452, 9003, 9453, 9748]
    children =
    - keys = [9456, 9459]
```

```
        parent keys = [9483, 9544, 9607, 9673, 9710]
        children = None
      - keys = [9484, 9486, 9509]
        parent keys = [9483, 9544, 9607, 9673, 9710]
        children = None
      - keys = [9558, 9574, 9585, 9593, 9596]
        parent keys = [9483, 9544, 9607, 9673, 9710]
        children = None
      - keys = [9620, 9625, 9634, 9635, 9662]
        parent keys = [9483, 9544, 9607, 9673, 9710]
        children = None
      - keys = [9676, 9686, 9692, 9696]
        parent keys = [9483, 9544, 9607, 9673, 9710]
        children = None
      - keys = [9732, 9742, 9747]
        parent keys = [9483, 9544, 9607, 9673, 9710]
        children = None
    - keys = [9841, 9868, 9887, 9949]
      parent keys = [8031, 8452, 9003, 9453, 9748]
      children =
      - keys = [9778, 9789, 9797, 9815, 9828]
        parent keys = [9841, 9868, 9887, 9949]
        children = None
      - keys = [9844, 9845, 9851, 9854, 9864]
        parent keys = [9841, 9868, 9887, 9949]
        children = None
      - keys = [9874, 9881]
        parent keys = [9841, 9868, 9887, 9949]
        children = None
      - keys = [9902, 9933, 9940]
        parent keys = [9841, 9868, 9887, 9949]
        children = None
      - keys = [9951, 9958, 9968, 9970, 9990]
        parent keys = [9841, 9868, 9887, 9949]
        children = None

---> Correct Solution:
- keys = [2906, 7608, 8860]
  children =
  - keys = [255, 631, 793, 1123, 1669]
    parent keys = [2906, 7608, 8860]
    children =
    - keys = [33, 79, 121, 183]
      parent keys = [255, 631, 793, 1123, 1669]
      children =
      - keys = [22, 25, 26, 31, 32]
        parent keys = [33, 79, 121, 183]
        children = None
      - keys = [40, 62]
```

```
      parent keys = [33, 79, 121, 183]
      children = None
    - keys = [81, 86, 113]
      parent keys = [33, 79, 121, 183]
      children = None
    - keys = [123, 127, 148, 173, 177]
      parent keys = [33, 79, 121, 183]
      children = None
    - keys = [192, 210, 226, 239, 251]
      parent keys = [33, 79, 121, 183]
      children = None
  - keys = [302, 404, 475, 504, 553]
    parent keys = [255, 631, 793, 1123, 1669]
    children =
    - keys = [267, 273, 277, 283, 300]
      parent keys = [302, 404, 475, 504, 553]
      children = None
    - keys = [318, 344, 358, 366]
      parent keys = [302, 404, 475, 504, 553]
      children = None
    - keys = [409, 421, 440, 458, 470]
      parent keys = [302, 404, 475, 504, 553]
      children = None
    - keys = [481, 484]
      parent keys = [302, 404, 475, 504, 553]
      children = None
    - keys = [518, 528, 532, 551]
      parent keys = [302, 404, 475, 504, 553]
      children = None
    - keys = [586, 594, 600, 619, 623]
      parent keys = [302, 404, 475, 504, 553]
      children = None
  - keys = [662, 686, 724, 728, 765]
    parent keys = [255, 631, 793, 1123, 1669]
    children =
    - keys = [635, 636, 641, 650]
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [672, 680, 681, 683, 684]
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [690, 694, 708, 719, 723]
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [725, 727]
      parent keys = [662, 686, 724, 728, 765]
      children = None
    - keys = [736, 749, 757]
      parent keys = [662, 686, 724, 728, 765]
```

```
      children = None
    - keys = [767, 768, 770, 776, 777]
      parent keys = [662, 686, 724, 728, 765]
      children = None
  - keys = [823, 863, 1007, 1071, 1091]
    parent keys = [255, 631, 793, 1123, 1669]
    children =
    - keys = [800, 813]
      parent keys = [823, 863, 1007, 1071, 1091]
      children = None
    - keys = [824, 826, 857]
      parent keys = [823, 863, 1007, 1071, 1091]
      children = None
    - keys = [873, 900, 912, 958, 993]
      parent keys = [823, 863, 1007, 1071, 1091]
      children = None
    - keys = [1019, 1020, 1029, 1053, 1064]
      parent keys = [823, 863, 1007, 1071, 1091]
      children = None
    - keys = [1074, 1075, 1087, 1090]
      parent keys = [823, 863, 1007, 1071, 1091]
      children = None
    - keys = [1099, 1104, 1109, 1115]
      parent keys = [823, 863, 1007, 1071, 1091]
      children = None
  - keys = [1133, 1199, 1214, 1452, 1489]
    parent keys = [255, 631, 793, 1123, 1669]
    children =
    - keys = [1126, 1127]
      parent keys = [1133, 1199, 1214, 1452, 1489]
      children = None
    - keys = [1138, 1152, 1153, 1173, 1192]
      parent keys = [1133, 1199, 1214, 1452, 1489]
      children = None
    - keys = [1207, 1212]
      parent keys = [1133, 1199, 1214, 1452, 1489]
      children = None
    - keys = [1222, 1283, 1299, 1337, 1367]
      parent keys = [1133, 1199, 1214, 1452, 1489]
      children = None
    - keys = [1479, 1485]
      parent keys = [1133, 1199, 1214, 1452, 1489]
      children = None
    - keys = [1508, 1523, 1532]
      parent keys = [1133, 1199, 1214, 1452, 1489]
      children = None
  - keys = [1909, 2050, 2383, 2465, 2601]
    parent keys = [255, 631, 793, 1123, 1669]
    children =
```

```
      - keys = [1683, 1757, 1777, 1862, 1889]
        parent keys = [1909, 2050, 2383, 2465, 2601]
        children = None
      - keys = [1960, 1984, 2031, 2046]
        parent keys = [1909, 2050, 2383, 2465, 2601]
        children = None
      - keys = [2098, 2111, 2150, 2210, 2323]
        parent keys = [1909, 2050, 2383, 2465, 2601]
        children = None
      - keys = [2410, 2462]
        parent keys = [1909, 2050, 2383, 2465, 2601]
        children = None
      - keys = [2483, 2520, 2588, 2591]
        parent keys = [1909, 2050, 2383, 2465, 2601]
        children = None
      - keys = [2625, 2714, 2751, 2767, 2833]
        parent keys = [1909, 2050, 2383, 2465, 2601]
        children = None
  - keys = [4262, 5517, 6104, 6384, 7074]
    parent keys = [2906, 7608, 8860]
    children =
    - keys = [3075, 3548, 4041]
      parent keys = [4262, 5517, 6104, 6384, 7074]
      children =
      - keys = [2957, 3035, 3040, 3055]
        parent keys = [3075, 3548, 4041]
        children = None
      - keys = [3177, 3351, 3418, 3486, 3545]
        parent keys = [3075, 3548, 4041]
        children = None
      - keys = [3645, 3793, 3920, 3964]
        parent keys = [3075, 3548, 4041]
        children = None
      - keys = [4056, 4142, 4143, 4201]
        parent keys = [3075, 3548, 4041]
        children = None
    - keys = [4392, 4656, 4854, 5303]
      parent keys = [4262, 5517, 6104, 6384, 7074]
      children =
      - keys = [4284, 4296, 4336, 4365]
        parent keys = [4392, 4656, 4854, 5303]
        children = None
      - keys = [4406, 4456, 4498, 4526, 4532]
        parent keys = [4392, 4656, 4854, 5303]
        children = None
      - keys = [4675, 4684, 4737, 4745, 4790]
        parent keys = [4392, 4656, 4854, 5303]
        children = None
      - keys = [5015, 5031, 5200, 5220, 5244]
```

```
      parent keys = [4392, 4656, 4854, 5303]
      children = None
    - keys = [5313, 5326, 5455, 5471, 5492]
      parent keys = [4392, 4656, 4854, 5303]
      children = None
  - keys = [5567, 5708, 5773, 5946, 6024]
    parent keys = [4262, 5517, 6104, 6384, 7074]
    children =
    - keys = [5549, 5551, 5557, 5558]
      parent keys = [5567, 5708, 5773, 5946, 6024]
      children = None
    - keys = [5609, 5610, 5615, 5631, 5636]
      parent keys = [5567, 5708, 5773, 5946, 6024]
      children = None
    - keys = [5711, 5713, 5766]
      parent keys = [5567, 5708, 5773, 5946, 6024]
      children = None
    - keys = [5827, 5837, 5858, 5899]
      parent keys = [5567, 5708, 5773, 5946, 6024]
      children = None
    - keys = [5949, 5974, 5996, 6003, 6007]
      parent keys = [5567, 5708, 5773, 5946, 6024]
      children = None
    - keys = [6025, 6040, 6061, 6067, 6102]
      parent keys = [5567, 5708, 5773, 5946, 6024]
      children = None
  - keys = [6213, 6320]
    parent keys = [4262, 5517, 6104, 6384, 7074]
    children =
    - keys = [6113, 6173, 6181, 6189]
      parent keys = [6213, 6320]
      children = None
    - keys = [6217, 6224, 6239, 6277, 6297]
      parent keys = [6213, 6320]
      children = None
    - keys = [6337, 6342, 6377, 6383]
      parent keys = [6213, 6320]
      children = None
  - keys = [6427, 6643, 6776, 6869, 7000]
    parent keys = [4262, 5517, 6104, 6384, 7074]
    children =
    - keys = [6392, 6401, 6409, 6421]
      parent keys = [6427, 6643, 6776, 6869, 7000]
      children = None
    - keys = [6522, 6541, 6579, 6595, 6604]
      parent keys = [6427, 6643, 6776, 6869, 7000]
      children = None
    - keys = [6676, 6704, 6711]
      parent keys = [6427, 6643, 6776, 6869, 7000]
```

```
      children = None
    - keys = [6816, 6854, 6856, 6865]
      parent keys = [6427, 6643, 6776, 6869, 7000]
      children = None
    - keys = [6876, 6877, 6904, 6920, 6982]
      parent keys = [6427, 6643, 6776, 6869, 7000]
      children = None
    - keys = [7032, 7043, 7046, 7070, 7073]
      parent keys = [6427, 6643, 6776, 6869, 7000]
      children = None
  - keys = [7125, 7221, 7348, 7512, 7539]
    parent keys = [4262, 5517, 6104, 6384, 7074]
    children =
    - keys = [7090, 7095]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7127, 7180, 7210, 7213]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7228, 7251, 7252, 7304, 7331]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7371, 7373, 7394, 7437, 7443]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7514, 7536]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
    - keys = [7541, 7596, 7600]
      parent keys = [7125, 7221, 7348, 7512, 7539]
      children = None
- keys = [8039, 8536]
  parent keys = [2906, 7608, 8860]
  children =
  - keys = [7651, 7741, 7807, 7934, 7980]
    parent keys = [8039, 8536]
    children =
    - keys = [7616, 7617, 7622]
      parent keys = [7651, 7741, 7807, 7934, 7980]
      children = None
    - keys = [7681, 7690, 7698, 7700, 7707]
      parent keys = [7651, 7741, 7807, 7934, 7980]
      children = None
    - keys = [7749, 7756, 7765]
      parent keys = [7651, 7741, 7807, 7934, 7980]
      children = None
    - keys = [7836, 7839, 7840, 7851, 7924]
      parent keys = [7651, 7741, 7807, 7934, 7980]
      children = None
```

```
      - keys = [7942, 7947, 7956]
        parent keys = [7651, 7741, 7807, 7934, 7980]
        children = None
      - keys = [7989, 8021, 8031]
        parent keys = [7651, 7741, 7807, 7934, 7980]
        children = None
    - keys = [8106, 8228, 8291, 8323, 8452]
      parent keys = [8039, 8536]
      children =
      - keys = [8040, 8086, 8088, 8091, 8105]
        parent keys = [8106, 8228, 8291, 8323, 8452]
        children = None
      - keys = [8145, 8179, 8189, 8190, 8204]
        parent keys = [8106, 8228, 8291, 8323, 8452]
        children = None
      - keys = [8244, 8271, 8277, 8282]
        parent keys = [8106, 8228, 8291, 8323, 8452]
        children = None
      - keys = [8293, 8301, 8304, 8314, 8316]
        parent keys = [8106, 8228, 8291, 8323, 8452]
        children = None
      - keys = [8340, 8351, 8388, 8391, 8419]
        parent keys = [8106, 8228, 8291, 8323, 8452]
        children = None
      - keys = [8473, 8474, 8475, 8513, 8528]
        parent keys = [8106, 8228, 8291, 8323, 8452]
        children = None
    - keys = [8657, 8725]
      parent keys = [8039, 8536]
      children =
      - keys = [8614, 8620, 8646, 8655]
        parent keys = [8657, 8725]
        children = None
      - keys = [8677, 8678, 8682, 8704, 8713]
        parent keys = [8657, 8725]
        children = None
      - keys = [8747, 8773, 8777, 8791, 8833]
        parent keys = [8657, 8725]
        children = None
  - keys = [9289, 9634, 9841]
    parent keys = [2906, 7608, 8860]
    children =
    - keys = [9012, 9163, 9254]
      parent keys = [9289, 9634, 9841]
      children =
      - keys = [8885, 8921, 8986, 8998, 9003]
        parent keys = [9012, 9163, 9254]
        children = None
      - keys = [9039, 9074, 9151]
```

```
      parent keys = [9012, 9163, 9254]
      children = None
    - keys = [9169, 9173, 9180]
      parent keys = [9012, 9163, 9254]
      children = None
    - keys = [9260, 9261, 9267, 9270, 9282]
      parent keys = [9012, 9163, 9254]
      children = None
  - keys = [9378, 9429, 9456, 9544, 9607]
    parent keys = [9289, 9634, 9841]
    children =
    - keys = [9310, 9313, 9314, 9353, 9368]
      parent keys = [9378, 9429, 9456, 9544, 9607]
      children = None
    - keys = [9397, 9404, 9406, 9411, 9418]
      parent keys = [9378, 9429, 9456, 9544, 9607]
      children = None
    - keys = [9431, 9437, 9445, 9448, 9453]
      parent keys = [9378, 9429, 9456, 9544, 9607]
      children = None
    - keys = [9459, 9483, 9484, 9486, 9509]
      parent keys = [9378, 9429, 9456, 9544, 9607]
      children = None
    - keys = [9558, 9574, 9585, 9593, 9596]
      parent keys = [9378, 9429, 9456, 9544, 9607]
      children = None
    - keys = [9620, 9625]
      parent keys = [9378, 9429, 9456, 9544, 9607]
      children = None
  - keys = [9676, 9696, 9748]
    parent keys = [9289, 9634, 9841]
    children =
    - keys = [9635, 9662, 9673]
      parent keys = [9676, 9696, 9748]
      children = None
    - keys = [9686, 9692]
      parent keys = [9676, 9696, 9748]
      children = None
    - keys = [9710, 9732, 9742, 9747]
      parent keys = [9676, 9696, 9748]
      children = None
    - keys = [9778, 9789, 9797, 9815, 9828]
      parent keys = [9676, 9696, 9748]
      children = None
  - keys = [9868, 9887, 9949]
    parent keys = [9289, 9634, 9841]
    children =
    - keys = [9844, 9845, 9851, 9854, 9864]
      parent keys = [9868, 9887, 9949]
```

```
        children = None
      - keys = [9874, 9881]
        parent keys = [9868, 9887, 9949]
        children = None
      - keys = [9902, 9933, 9940]
        parent keys = [9868, 9887, 9949]
        children = None
      - keys = [9951, 9958, 9968, 9970, 9990]
        parent keys = [9868, 9887, 9949]
        children = None

Note: Once a Sub-Test fails, no more Sub-Tests in that Test Set are run
Test Set 8: Failed!
```

9) test_9 (test_generated_cases.simplebtree) (0/10)

```
Test Set 9: Performing 5 tests with size = 1000 and type = insert and d
Sub-Test #1: Passed (Details Follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,9
insert,5380
insert,2943
.
.
.
insert,3896
insert,4197
search,3790
---> Your Solution:
[2]

---> Correct Solution:
[2]

Sub-Test #2: Failed (Details follow)
---> Tracefile:
NOTE: TRACEFILE ABBREVIATED DUE TO SIZE!
initialize,6
insert,6088
insert,8403
.
.
.
insert,5674
insert,9511
search,5097
---> Your Solution:
[2, 0]

---> Correct Solution:
[2]

Note: Once a Sub-Test fails, no more Sub-Tests in that Test Set are run
Test Set 9: Failed!

Test Failed: False is not true
```

C3

● Ungraded

Student

Katherine Alvarenga

Total Points

- / 100 pts

Autograder Score

70.0 / 100.0

Failed Tests

5) test_5 (test_generated_cases.simplebtree) (0/10)

8) test_8 (test_generated_cases.simplebtree) (0/10)

9) test_9 (test_generated_cases.simplebtree) (0/10)

Passed Tests

0.0) test_submitted_files (test_file_exists.TestFiles) (10/10)

1) test_1 (test_generated_cases.simplebtree) (10/10)

2) test_2 (test_generated_cases.simplebtree) (10/10)

3) test_3 (test_generated_cases.simplebtree) (10/10)

4) test_4 (test_generated_cases.simplebtree) (10/10)

6) test_6 (test_generated_cases.simplebtree) (10/10)

7) test_7 (test_generated_cases.simplebtree) (10/10)